

The Oxford Handbook of Quantitative Methods in Psychology, Vol. 1

Todd D. Little (ed.)

<https://doi.org/10.1093/oxfordhb/9780199934874.001.0001>

Published: 2013

Online ISBN: 9780199971299

Print ISBN: 9780199934874

CHAPTER

8 The IRT Tradition and its Applications

R.J. de Ayala

<https://doi.org/10.1093/oxfordhb/9780199934874.013.0008> Pages 144–169

Published: 01 October 2013

Abstract

Item response theory (IRT) is based on the premise that one or more unobservable (latent) variables are manifested in observable behaviors. These discrete observable behaviors are converted into continuous measurements through the application of an IRT model. We present and discuss our IRT models in terms of frames of reference, psychometric purpose, and type of response data. The models presented are applicable for affective, attitudinal, and proficiency data. The benefits and advantages of IRT models are given. We briefly discuss parameter estimation and provide a nonexhaustive list of estimation programs. The processes of model–data fit are presented as are transformation of our continuum’s metric.

Keywords: Item response theory, latent variable, logistic model, normal ogive model, psychometrics, maximum likelihood

Subject: Research Methods in Psychology, Psychology

Series: Oxford Library of Psychology

The Item Response Theory Tradition and Its Applications

In this chapter I discuss a theory of item responses. This paradigm, item response theory (IRT), posits the existence of one or more unobservable (latent) variables that are manifested in observable behaviors. Our construct(s) of interest is represented by the latent variable(s), whereas the observable behaviors may be an individual’s responses to items or they may be the observations of (expert/knowledgeable) judges/raters of an individual’s behavior. As such, the term *item response* can reflect an individual’s response to a question from an attitude or affective scale, a vocational inventory, a proficiency examination, or it may be a judge’s rating.

These different ways in which our item responses arise reflect different reference frames. In the typical case, we have persons responding to items or a two–facet frame of reference. As an example, we might have patient responses to a quality–of–life inventory or examinee responses to test questions. In contrast, there are assessment situations that involve more than two facets. For example, clinicians’ judgments of patients’

responses would be a three-facet frame of reference (i.e., patients by responses by clinicians). Stated another way, we may apply IRT in two-facet situations or in cases that have more than two-facet cases, such as to person by items by judges data (i.e., three facets).

The two- and three-facet reference frames differ in that with the two-facet we can directly measure an individual on a latent variable without using an intermediary, such as a judge or a rater. Typically, the three-facet reference frame uses a judge or rater as an intervening agent. As will be seen below, the IRT models used in the three-facet reference frame are modified versions of those used in the two-facet frame of reference.

p. 145 We can further modify our models to address different psychometric objectives. Broadly speaking, if we are interested in locating people (or items) on a latent variable, then we are *describing* the respondents (or items) in terms of our measurements. In this case, our models might be referred to as *descriptive IRT models* (e.g., Wilson & De Boeck, 2004). However, a second objective could be to *predict* or *explain* the latent item and/or person variables from manifest item and person characteristics. In this case, these models are known as *explanatory IRT models* (e.g., Wilson & De Boeck, 2004). This may occur when we are interested in testing, for example, a theory of cognitive development by making theory-based predictions about respondent locations. As such, the individual respondent is not necessarily the sole focus of the instrument's administration. These two objectives, describing and predicting, can be combined in our models. For example, we may be interested in assessing respondents' efficacy for weight loss and also be interested in explaining this efficacy in terms of respondent characteristics.

The foregoing is intended to show that IRT is more broadly applicable than may be the reader's impression from the literature. In short, when we apply IRT for measurement, we are not restricted to a two-facet frame of reference nor are we forced to confine ourselves to simply describing a person or item's location on a continuum. In the following we will, for simplicity, address the common situation of a two-facet reference frame as well as a descriptive psychometric objective. We begin with some benefits of IRT, progress to a general model, and then proceed to specific IRT models. Following these models we briefly discuss the principles of estimation, model assumptions, fit analysis, metric transformations, and sample sizes. Table 8.1 contains a listing of commonly used symbols used in this chapter, whereas Table 8.2 is a glossary of

p. 146 commonly used terms. ↵

Table 8.1. Commonly Used Symbols

Symbol	Comment
$p_i(x_i)$	Probability of a response of x on item i
Φ	The cumulative distribution function of the unit normal distribution
Ψ	The logistic distribution function
D	A scaling constant equal to 1.702
γ_i	Intercept parameter for item i 's logit regression line
α_i	Slope parameter for item i 's logit regression line; item discrimination
$\hat{\alpha}_i$	Estimated slope for item i 's logit regression line (discrimination)
δ_i	Item i 's location parameter on the latent construct; $\delta_i = \alpha_i / \gamma_i$
$\hat{\delta}_i$	Item i 's estimated location
θ_r	Person r 's location on the latent construct
$\hat{\theta}_r$	Person r 's estimated location on the latent construct
$I_i(\theta)$	Item information
$I(\theta)$	Total (or test) information
m	The number of categories
m	The number of transitions between categories
τ_k	The transition from rating category $k - 1$ to rating category k on an item with $k = 1 \dots m$; a.k.a., threshold parameter
π_v	Latent class V 's proportion
$\zeta_{,k}$	Metric transformation coefficients

Table 8.2. Commonly Used Terms

Term	Acronym/Symbol	Definition
Item response function (item characteristic curve)	IRF or ICC	The probability of a response of 1 as function of item and person parameters
Option response function	ORF	The probability of responding in or selecting a particular item option as function of item and person parameters
Item information	$I_i(\theta)$	The reduction in uncertainty about a person's location provided by an item
Total (test) information	$I(\theta)$	The reduction in uncertainty about a person's location provided by an instrument
<i>Estimation</i>		
Likelihood function	L	The probability of a set of observations as a function of unknown parameter(s)
Log-likelihood function	$\ln L$	Logarithmic transformation of the likelihood function
Maximum likelihood estimation	MLE	A parameter estimation technique in which the location of the likelihood's maximum is the estimate of the unknown parameter(s) underlying the likelihood function
Maximum <i>a posteriori</i>	MAP	A Bayesian parameter estimation technique in which the mode of the posterior distribution of the likelihood is the estimate of the unknown parameter(s) underlying the likelihood function
Expected <i>a posteriori</i>	EAP	A Bayesian parameter estimation technique in which the mean of the posterior distribution of the likelihood is the estimate of the unknown parameter(s) underlying the likelihood function
<i>IRT model assumptions</i>		
Conditional independence (local independence)		For any group of individuals that are characterized by the same latent location(s) the conditional distributions of the item responses are all independent of one another
Functional form		The data follow the function specified by the model
Dimensionality		Observations on the manifest variables are a function of one or more continuous latent person variables

Invariance		The estimate's characteristic of not changing (in a relative sense) across different samples
Differential item functioning	DIF	An item that displays different statistical properties for different manifest groups after the groups have been matched on a measure of the construct
Focal group		A manifest group of respondents that is investigated to see whether it is disadvantaged by an item
Reference group		A manifest group of respondents that is used as the comparison group to see if the focal group is disadvantaged by an item
Linking		The alignment of two metrics with one another

p. 147 Benefits of Item Response Theory

With IRT models, it is possible to design instruments with specific characteristics. As an example, we may desire to create an instrument that provides maximum accuracy in person estimation at a particular decision point on the latent continuum (e.g., a cut point). Alternatively, we might need an instrument that provides equiprecise estimates across the continuum. In both cases, we select items not only for appropriate content coverage, but also for their parameter estimates that achieve our objective. Moreover, depending on the IRT model, it is possible to design items that we believe are consistent with cognitive theory and test whether they are.

Item response theory has several advantages over Classical Test Theory (CTT). For example, with CTT a person's observed score is directly related to the instrument's characteristics. This is easiest to see in the context of proficiency testing. For example, if one administers a difficult exam to a group of examinees, then their observed scores will be systematically less than the scores they would have received had they been administered an easy exam. In contrast, IRT person estimates are independent of the specific sample of items administered to the person (i.e., "item-free" person estimation). It is this property that allows *computerized adaptive testing* (CAT) to tailor tests to individual examinees and yield person location estimates that can be compared to one another. (For more information on CAT, see Drasgow & Olson-Buchanan, 1999; Parshall, Spray, Kalohn, & Davey, 2002; Reckase, 1989; Sands, Waters, & McBride, 1997; and van der Linden & Glass, 2010.)

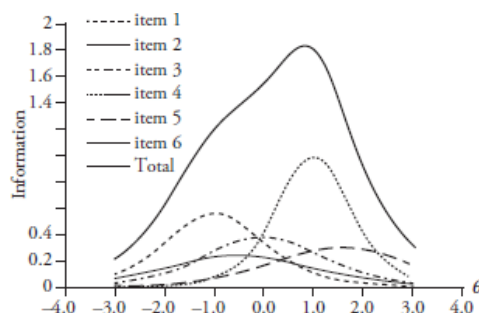
A second advantage of IRT is item parameter estimates that are not dependent on the particular sample of examinees (i.e., "person-free" item estimation). In contrast, traditional item statistics, such as item difficulty (i.e., proportion correct) and item discrimination (e.g., the point biserial), depend on the examinee sample. Again this is easiest to see in the context of proficiency testing. For example, an item administered to high-ability examinees will show a higher item difficulty (i.e., an easy item) than when administered to low ability examinees.

Generally, "person-free estimation of item parameters" and "item-free estimation of persons" are examples of item parameter and person parameter *invariance*, respectively. Therefore, it is possible to create instruments that are free of the particular respondents used in obtaining item parameter estimates as well as obtaining person location estimates that transcend the particular instruments used in the assessment provided that one has model-data fit.

The third advantage of IRT over CTT concerns measurement error. In CTT, one's assessment of the measurement error for an instrument (i.e., the standard error of measurement) is constant for all persons regardless of his or her observed score; it also depends on the individuals to whom the instrument is administered. However, we know that the amount of measurement error varies across the observed score scale (see Haertel, 2006). As such, the standard error of measurement overestimates the amount of measurement error in some observed scores while underestimating the degree of error in other observed scores. In contrast, with IRT we have assessment of measurement error for each person rather than this aggregated measurement error. This measurement error statistic, the standard error of estimate (SEE), provides us with an index of the accuracy for each of our person location estimates ($\hat{\theta}$).

A person's SEE indicates how uncertain we are about the person location estimate. The larger the SEE, the less certain we are about where the person is located. Conversely, a small SEE means that our instrument is providing us with lots of information about the person's location. This concept of the information that an item and an instrument have for estimating person locations is not found in CTT. In IRT, information is defined at both the item- and instrument-levels. *Item information*, $I_i(\theta)$, refers to the amount of information an item provides for estimating a person's location. Moreover, one can sum the individual item information across the L items on an instrument to produce its *total (or test) information*, $I(\theta)$; that is, $I(\theta) = \sum^L I_i(\theta)$; note the use of subscript i to reflect item information.

As an example, Figure 8.1 shows the item information and total information functions for a five-item social anxiety instrument. The total information function (solid bold) shows that our instrument provides the most accurate person location estimates (i.e., has the most information) around 1, but the instrument is useful for accurately estimating persons from roughly -2 to 2.5 . Stated another way, the location of an instrument's maximum information is also where the person location estimates have the smallest SEEs, because information is inversely related to the square of the SEE. The figure also shows the item information functions for each of the five items that make up the total information function. These items provide their respective maximum information at different locations across our scale and in different amounts; an item's maximum information has a direct relationship to how well an item discriminates among respondents. For example, item 4 provides its peak information at 1 and over a comparatively narrower range than does item 3. This property allows us to combine items that have different item information maxima and distributions to design an instrument that has specific estimation properties (e.g., equiprecise estimates throughout the continuum). Thus, the information function can be more useful in assessing an instrument's psychometric quality than reliability estimates. As stated above, these benefits and advantages can only be realized when we have model-data fit. Moreover, implied in some of the advantages is a "degree of reasonableness." For example, with item-free person estimation we assume that the different samples of items come from an item pool that measures the same construct or that with person-free estimation of item parameters our examinee samples come from the same population. Furthermore, IRT is not a panacea for poorly designed or worded items. However, when standard psychometric principles are followed, the use of IRT will provide benefits over CTT.

Figure 8.1:

Total and item information plot for a five-item instrument.

A General Model

We begin with a general formulation for an IRT model

$$p_i(x_i) = f(\Xi), \quad (1)$$

where the probability, p_i , of a response x to item i is a function, $f(\bullet)$, of the item and person parameters represented by Ξ ; Ξ is the Greek letter Xi. The specific nature of Equation 1 depends on the psychometric context.

One psychometric context concerns our objective in administering our instrument. As mentioned above, this psychometric objective can be to (1) describe the respondents and/or items, (2) predict or explain the latent person and/or item variables from manifest item and person characteristics, or (3) some combination of description and prediction (or explanation).

A second psychometric context concerns the responses, x . We categorize our responses as polytomous (e.g., rating scale, Likert scale) or dichotomous (e.g., True/False, correct/incorrect). In this latter case, we will not be concerned with whether these responses arose, for example, from the dichotomization of a normally distributed response variable (cf. tetrachoric correlation). As such, we can classify our IRT models as those used for polytomous data and those used for dichotomous responses. Both polytomous and dichotomous IRT models may be used together to obtain parameter estimates for an assessment. Moreover, specific dichotomous models are constrained variants of specific polytomous models. Note that the responses that we model may be (1) directly provided by a respondent (e.g., responses to a Likert scale, binary responses), (2) assigned to a respondent by a judge/rater according to a rubric, or (3) the outcome of scoring the responses (e.g., correct or incorrect). The application of our IRT model typically translates the discrete responses into a continuous measurement of the respondents and items. It may be obvious that these two psychometric contexts can co-occur (i.e., describing respondents when our responses are polytomous).

We now turn to the issue of the nature of the function $f(\bullet)$. To discuss this we need to adopt a historical perspective; in the following, assume that we have binary responses. Over the previous century, several prominent psychometricians have used the standard cumulative normal distribution (i.e., the normal ogive) as a model for working with item responses (e.g., Lord, 1952; Thurstone, 1925; Tucker, 1946). As such, the function $f(\bullet)$ can be defined as a probit link function between the probability of the response x to an item ($p(x)$) and Ξ . That is,

$$p(x) = \Phi(\Xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\Xi} \exp\left(-\frac{1}{2}z^2\right) dz, \quad (2)$$

p. 149 where Φ is the cumulative distribution function of the unit normal distribution (i.e., $f(\Xi) = f_{\text{probit}}(\Xi) = \Phi(\Xi)$) and x is a dichotomous response variable (i.e., $x = \{0, 1\}$). Equation 2 is a linear normal probability unit or probit model; some refer to probit as normit to reflect the use of a *normal* probability unit.

An alternative candidate for the function $f(\bullet)$ is the (inverse) logit link function between the probability of the response and Ξ

$$p(x) = \Psi(\Xi) = \frac{1}{1+e^{-\Xi}} = \frac{e^{\Xi}}{1+e^{\Xi}}, \quad (3)$$

where Ψ is the logistic distribution function (i.e., $f(\Xi) = f_{\text{logit}}(\Xi) = \Psi(\Xi)$). Equation 3 is a linear logistic probability unit or logit model. In some cases, it is convenient and/or instructive to represent Equation 3 in a log-odds format¹ (also known as the logit transformation):

$$\ln\left(\frac{p(x)}{1-p(x)}\right) = \Xi \quad (4)$$

To summarize, there are IRT models that use the normal ogive to model the item response function, whereas others use the logistic function. Because the normal ogive models predate the logistic models, we sometimes see the logistic models written with a scaling constant, D , to maximize the similarity of the logistic results with those from the corresponding normal ogive model results (i.e., $\text{probit} = \text{logit} * D = \text{logit} * 1.702$); de Ayala (2009) contains more information on D .

Given that we can transform the logistic class model results to those of the normal ogive class of models (and vice versa), the decision between the two is based on pragmatic considerations, such as available software. Moreover, whether we use the logistic models (with or without the D scaling constant) or the normal ogive models does not affect our model-data fit. However, because the logistic model class does not require integration, they are more commonly used than the normal ogive models. When we use the normal ogive models (or the logistic models with the D scaling constant), our results are said to be on the *normal metric*, otherwise the results are on the *logistic metric*. The importance of this distinction comes into play when making comparison with other related techniques. For example, because there is fundamentally no difference between a single-factor analytic model and a unidimensional IRT model, it is possible to estimate IRT item parameters using a factor analysis routine. However, because these results are on the normal metric, to compare them or to use them with estimates from the logistic class of models requires converting from the normal metric to the logistic metric or vice versa.

two-parameter model

Let us now turn our attention to Ξ and present our first IRT model, the *two-parameter* model. We begin with this model because we consider it to be the *nexus model*. That is, almost all other models are an extension or a constrained version of the two-parameter model. For the two-parameter model we let

$$\Xi = \gamma_i + \alpha_i \theta_r, \quad (5)$$

where γ_i and α_i are the intercept and slope parameters for item i 's logit regression line, respectively, and θ_r is person r 's location on the latent construct. We will refer to Equation 5 as either the *slope-intercept* or the *linearized* form. The two-parameter model is so-called because it contains two parameters to characterize the item; we discuss these parameters below.

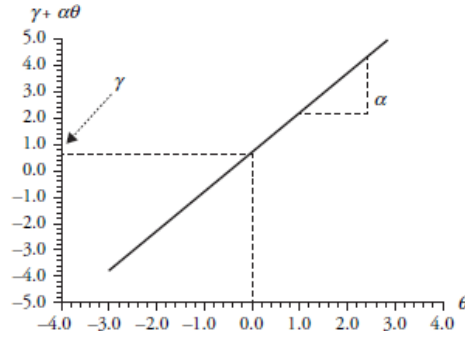
If we substitute Equation 5 into Equation 3, then we obtain the *two-parameter logistic* (2PL) model. The 2PL model states that the probability of a response of 1 given person r 's latent location of θ is given by

$$\begin{aligned} p_i(x_i = 1|\theta_r) &= \Psi(\gamma_i + \alpha_i \theta_r) \\ &= \frac{1}{1 + e^{-(\gamma_i + \alpha_i \theta_r)}} = \frac{e^{\gamma_i + \alpha_i \theta_r}}{1 + e^{\gamma_i + \alpha_i \theta_r}}. \end{aligned} \quad (6)$$

By a “response of 1” we mean that the response x has been categorized or coded as a 1 and may represent, for example, a correct response, a response of True, and so on. A “response of 0” would represent the complementary event, such as an incorrect response, a response of False, and so forth.

The theoretical range of θ_r, α_i , and γ_i is $-\infty$ to ∞ . Empirical values for θ_r typically fall between -3 and 3 , with respondents located toward the upper end of the continuum (e.g., $\theta_r = 2$) reflecting more of whatever the latent variable is than respondents located toward the lower end of the continuum (e.g., $\theta_r = -2$). Items that discriminate well have values of α_i above 0.8; a negative α_i indicates an item that is inconsistent with the model or has been coded incorrectly. As will be seen below, the observed range of γ_i is not as important as that of a related parameter, the item's location.

To help us understand the meaning of γ_i and α_i , we present the logit, $\gamma + \alpha\theta$, as a function of θ in Figure 8.2. Let us assume that the example item shown in this figure is from a personality inventory having to do with social anxiety and uses a true/false response format. Stated another way, our latent variable of interest is social anxiety. Moreover, the right side of the continuum reflects higher social anxiety than does the left side of the continuum. To make our example more concrete, our item is “I feel socially anxious at parties.” Given our true/false response format, we code a response of true as a 1 and a response of 0 reflects a response of false. The item's logistic regression line has a slope (α) of 1.5 and an intercept (γ) equal to 0.75. As can be seen, the item's intercept parameter is the point on the logit scale ($\gamma + \alpha\theta$) where the logit regression line intersects with the vertical axis (ordinate) when $\theta = 0$ and the slope of the line (indicated by the right triangle) is α .

Figure 8.2:

Logit space plot for an item with $\alpha = 1.5$ and $\gamma = 0.75$.

The logistic regression line shows that the log odds (or logit) of a true response increases as θ increases. For example, assume a person is located at 1 (i.e., $\theta = 1$). Starting at $\theta = 1$ on the horizontal axis (abscissa), going up to the logit regression line, and then projecting over to the ordinate, we see that the logit is 2.25. That is, a person located at 1 has a log odds of 2.25 (logit = 2.25) of responding true to our item. In terms of *odds*, a person located at 1.0 is almost 9.5 times more likely to respond true to being socially anxious at parties rather than false (i.e., $\text{odds} = \exp(2.25) = 9.49$). Conversely, a person located at -1 has a logit value of -0.75 or odds of responding true to our item of 0.47. In other words, a person located at -1 is more than twice as likely to respond false to being socially anxious at parties as oppose to answering true ($1/.47 = 2.12$).

We can reparameterize Equation 5 into a *difference* or *deviate* form by letting $\gamma_i = -\alpha_i \delta_i$. In this parameterization, δ_i is item i 's *location* on the latent construct. Furthermore, this reparameterization permits another way of interpreting γ_i —namely, as the interaction of an item's discrimination and its location. If we substitute $-\alpha_i \delta_i$ for γ_i in Equation 5 and factor, then we obtain the *logistic deviate* form of Equation 5,

$$\Xi = \alpha_i (\theta_r - \delta_i). \quad (7)$$

In this form we see that the primary determinant of the probability of a response of 1 is the weighted difference between the person and item locations. As such, persons and items are located on the same latent continuum. The theoretical range of δ_i is $-\infty$ to ∞ with empirical values for δ_i typically falling between -3 and 3 logits. Generally speaking, items with δ_i 's greater than 0 indicate comparatively more difficult to endorse items than items located below 0. In the context of proficiency assessment, the item latent location parameter δ is referred to as *item difficulty* and the person latent location parameter (θ) is known as *ability* or *proficiency*.

Applying Equation 7 to our logistic distribution function, $\Psi(\Xi)$ (Equation 3), we obtain an alternative representation of the 2PL model (all other terms are essentially the same as above):

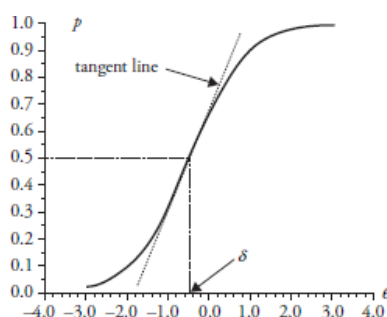
$$p_i(x_i = 1 | \theta_r, \alpha_i, \delta_i) = \Psi(\alpha_i (\theta_r - \delta_i)) = \frac{1}{1 + e^{-\alpha_i (\theta_r - \delta_i)}} = \frac{e^{\alpha_i (\theta_r - \delta_i)}}{1 + e^{\alpha_i (\theta_r - \delta_i)}}. \quad (8)$$

Equation 8 has a graphical analog to the logit space plot shown in Equation 6. Specifically, the graphical representation of the relationship between θ and the probability of a response of 1 for an item is the (predicted) *item response function* (IRF); sometimes the IRF is called the *item characteristic curve* (ICC). Figure 8.3 contains the IRF for our example social anxiety item. As can be seen, as θ increases, so does the probability of a response of 1.

To obtain the IRF corresponding to our social anxiety item ($\alpha_i = 1.5, \gamma_i = 0.75$), we first determine that the item is located at $\delta_i = -\alpha_i / \gamma_i = -1.5 / 0.75 = -0.50$. We then calculate the probability of a response of 1 according to our IRT model (e.g., Equation 8) for values of θ from -3 to 3 . For our item, we see that respondents located above -0.5 on the latent continuum are most likely to respond true to feeling anxious at parties and that persons located below -0.5 are most likely to respond false. In fact, a person located at -2 or below has less than 10% chance of responding true, whereas a person located at 1 or above has at least a 90% chance of responding true (i.e., given $\theta = 1, \alpha_i = 1.5$, and $\delta_i = -0.50$, then according to Equation 8 the probability of answering true is 0.905). In short, the more socially anxious our respondent is (i.e., the higher his or her θ), the more likely it is that he or she will respond true to our item.

p. 151 There are other features of the IRF shown in Figure 8.3 worth making salient. First, and as stated above, the item location δ is on the same continuum as the respondent location θ . This characteristic of having items and people located on the same continuum is not found in CTT. (In CTT an item's difficulty is on a 0 to 1 scale and the observed score is not.) Further, it can be seen that δ corresponds to the inflexion point for the IRT and that the probability of a response of 1 at this point is 0.50 . In this context and at first glance it may be difficult to visualize how the item's other parameter, α_i , comes into play. As was the case with the slope-intercept form of the 2PL model α_i is related to the slope, but in Equation 8 it is proportional to the slope of a line tangent to the IRF at δ . Because the slope of the line relates to how well the item can differentiate among respondents located at different points on the continuum, α is typically referred to as the item's *discrimination* parameter. As one would expect given the use of the logistic function, the lower and upper asymptotes of the IRF are 0 and 1, respectively.²

Figure 8.3:



IRF for an item with $\alpha_i = 1.5$ and $\delta_i = -0.5$.

one-parameter model

Equation 8 can be simplified by imposing the constraint that all items on a scale share a common discrimination parameter. Therefore, items differ from one another only in terms of their locations on the latent continuum. Concerning ourselves solely with the logistic deviate, this constraint would be represented by dropping the subscript on α . Thus, we have

$$\Xi = \alpha (\theta_r - \delta_i). \quad (9)$$

Typically, the one-parameter model is expressed using logistic distribution function, $\Psi(\Xi)$. In this case the model is called the *one-parameter logistic* (1PL) model.

The implications of imposing a constant α constraint are that the manifest observed score, X_r , is a sufficient statistic for estimating a person r 's location and the sum of item responses across respondents (i.e., the manifest item score, $q_i = \sum_r x_{ri}$) is a sufficient statistic for estimating the item i 's location. As such, all persons with the same observed score obtain the same estimated location, $\hat{\theta}$, and all items with the same item score have the same estimated location, $\hat{\delta}$. This characteristic can be used to simplify parameter estimation and facilitates communicating the results to laypeople because of the direct relationship between the manifest and latent variables. The 1PL model is sometimes referred to as the Rasch model (Rasch, 1961, 1980), although others restrict the equivalence of the 1PL and the Rasch model to when $\alpha = 1$.

extending our model

Rather than concerning ourselves only with locating respondents on the latent continuum, we may wish to predict or explain the differences between respondents in terms of their person parameters. In this context, our model could be considered a *person explanatory* model (see Wilson & De Boeck, 2004). This prediction or explanation is based on a weighted linear composite of manifest variables—that is, $\theta_r = \sum b_k Z_k + \varepsilon_r$. As an example, assume that we believe that our respondents' social anxiety (i.e., their locations on the social anxiety continuum) can be “explained” in terms of two predictors, the respondent's experience with past public humiliations (a binary variable, yes/no), and tendency to worry. Our analysis would allow us to determine the effect of each predictor in explaining the variability of social anxiety (latent) locations as well as an assessment of how much of the variability is accounted for/explained.

linear logistic test model

We can modify the logistic deviate used with the 1PL/Rasch model to incorporate information about the cognitive operations that underlie our observed responses. The resulting model is the linear logistic test model (LLTM). The LLTM (Fischer, 1973) is an example of the aforementioned psychometric objective of predicting or explaining the latent item variable from manifest item characteristics. Again, concerning ourselves solely with the deviate we would have

$$\Xi = \alpha (\theta_r - \delta_i) = \alpha \left(\theta_r - \left[\sum_s f_{is} \eta_s + C \right] \right), \quad (10)$$

p. 152 where the item's location is a weighted linear composite of item characteristics—that is, $\delta_i = \sum_s f_{is} \eta_s + C$. The η_s is a basic parameter associated with elementary component s ($s = 1 \dots S$), f_{is} is the weight of component s for item i , and C is a normalization constant. The f_{is} s could be the hypothetical frequencies with which each component s influences the response to item i or may simply reflect whether a component is necessary for responding to an item; when S equals the number of items, then the LLTM is equivalent to the Rasch model (Embretson, 1984). The η_s s typically reflect the psychological structure of an item. For example, they may correspond to cognitive operations underlying a response (or the difficulties

thereof), instructional conditions (characterized by their efficacy), item characteristics determined by an experimental design, and so on. Typically, $\alpha = 1$ in the LLTM.

The LLTM is another example of a model that fulfills the objective of predicting or explaining the latent item variables from manifest item characteristics. If the data used with the LLTM arise from an experimental design that investigates or that uses item characteristics to explicate the response data, then we view the LLTM as serving an explanatory objective. However, if the data arise from a nonexperimental setting (i.e., without experimental control and random selection and assignment), then it is most accurate to consider the LLTM as fulfilling a predictive objective. The LLTM can be considered an example of an *item explanatory* model (see Wilson & De Boeck, 2004).

Although all occurrences of the LLTM of which we are aware formulate the LLTM using $\Psi(\Xi)$ (Equation 3), the model could utilize a probit link (i.e., the *linear probit test model*). The reader is referred to Baker (1993a), Embretson (1985, 1996), Fischer (1973), Frederiksen, Mislevy, and Bejar (1993), and Irvine and Kyllonen (2002) for greater detail on the LLTM and its application.

facet model

We started this chapter talking about frames of reference. One of our scenarios was a three-facet reference frame involving clinicians' judgments of patients' conditions based on the patients' responses. In this case, our manifest observations arise or are the product of the interaction of the patient (facet 1), the responses (facet 2), and the clinician (facet 3). Therefore, the observations that are the basis of our measurement are given by a clinician's judgment of a patient's responses rather than directly from the patient's responses. As stated above, a key characteristic of a three-facet reference frame is that the data come from an *intervening* agent's judgments of an individual's interactions with the questions that are posed to him or her. In contrast, in a two-facet reference frame, our data arise *directly* from our respondent's interactions with the items.

We can extend the 1PL/Rasch model from a two-facet frame of reference (i.e., participants by items) to a multifacet frame of reference. This extension is known as the *Facet model* or the *Many-Facet Rasch model* (MFRM; Linacre, 1988, 1989). In contrast with the above models' focus on dichotomous responses, the MFRM can be used with polytomous ratings as well as dichotomous responses. Our introduction of modeling polytomous responses reflects the fact that, typically, judges/raters use a rating scale with more than just two categories (i.e., a polytomous rating scale); implicit in a rating scale is that the rating response categories are ordered.

Because the MFRM can theoretically be applied to any number of facets, to present it we need to specify the number of facets. In this light the logistic deviate for the MFRM for a (most common) three-facet framework is given by

$$\Xi = \alpha(\theta_r - \delta_i - \omega_j - \tau_k), \quad (11)$$

where α and δ_i represent one facet (item i) and are defined as above,

θ_r represents a second facet (respondent r) and is defined as above, ω_j represents the third facet (the j^{th} rater/judge/grader), τ_k represents the transition from rating category $k - 1$ to rating category k on an item and $k = 1 \dots m$; the non italicized "m" is the number of transitions between categories.

The symbol ω_j represents the j^{th} rater/judge/ grader's severity, whereas τ_k represents the relative difficulty of being rated in the k^{th} category over the $k^{th} - 1$ category (e.g., category 1 vs. category 0). The lowest rating category is coded 0 and so the number of *rating or response categories* is given by $m (= m + 1)$; we *italicize* "m" to represent the number of categories. (In the context of dichotomous models, our items would have two response categories ($x = \{0, 1\}$) and m would equal 2 with one transition between the response categories (i.e., $m = 1$) that occurs at the item's location, δ_i .) The three-facet deviate shown in Equation 11 can be extended to include additional facets (e.g., occasions) by adding a corresponding parameter to the deviate. For an example of a four-facet reference frame, see Smith and Kulikowich (2004).

p. 153 In contrast to commonly used logistic distribution function presentation of an IRT model, the MFRM is typically represented in the log-odds format (Equation 4). This form is obtained by substituting Equation 11 into Equation 4. Thus, the log-odds of respondent r being given a rating in category k instead of a rating in category $(k - 1)$ on item i by judge j is

$$\ln \left(\frac{p(x_{rijk})}{p(x_{rij(k-1)})} \right) = \Xi = \alpha (\theta_r - \delta_i - \omega_j - \tau_k), \quad (12)$$

where

$p(x_{rijk})$ is the probability of respondent r being given a rating in category k , on item i by judge j , and $p(x_{rij(k-1)})$ is the probability of receiving a rating in category $(k - 1)$ on item i by judge j , and $k = \{1, \dots, m\}$.

In terms of our logistic distribution function, $\Psi(\Xi)$ (Equation 3), our MFRM would be written as

$$\begin{aligned} p(x_{rijk} | \theta_r, \alpha, \delta_i, \omega_j, \tau_k) &= \Psi(\Xi) \\ &= \frac{e^{\sum_{k=0}^x \alpha(\theta_r - \delta_i - \omega_j - \tau_k)}}{1 + \sum_{v=1}^m e^{\sum_{k=0}^x \alpha(\theta_r - \delta_i - \omega_j - \tau_k)}} \\ &= \frac{e^{\sum_{k=0}^x \alpha(\theta_r - \delta_i - \omega_j - \tau_k)}}{\sum_{v=0}^m e^{\sum_{k=0}^x \alpha(\theta_r - \delta_i - \omega_j - \tau_k)}}, \end{aligned} \quad (13)$$

where $p(x_{rijk})$ is the probability that respondent r is judged by rater j to be in item i 's category k (i.e., the probability of a rating score of x_{rijk} where $x_{rijk} = \{0, 1, \dots, m\}$).

As presented above, the MFRM assumes that the rating scale is constant across an item set as well as for all judges/raters/graders; these assumptions may be relaxed. Moreover, as was the case with the LLTM, α is typically set to 1. A MFRM analysis produces estimates of the person and item locations as well as an assessment of the judges' severity.

generalized partial credit and partial credit models

As was the case with the MFRM, the *generalized partial credit* (GPC) model (Muraki, 1992) can be used with ordered polytomous response data. As an example, consider the National Survey of Student Engagement in which students respond to a series of questions designed to measure collegiate quality using a response format such as “very often,” “often,” “sometimes,” and “never.” Alternatively, the ordered polytomous response data can represent a partial credit proficiency assessment situation (i.e., 0 points = no credit, 1 point = partial credit response, 2 points = full credit) or a Likert response scale.

The GPC model assumes that the probability of selecting a particular response category over the previous one is governed by the dichotomous 2PL model (Equation 8). As a result of applying this “dichotomized process” across an item’s successive response categories, one obtains a model whose logistic deviate is:

$$\Xi = \sum_{h=1}^k \alpha_i (\theta_r - \delta_{ih}). \quad (14)$$

Therefore, the GPC model is

$$p(x_{ik} | \theta_r, \alpha_i, \delta_i) = \frac{\exp \left[\sum_{h=1}^k \alpha_i (\theta_r - \delta_{ih}) \right]}{\sum_{c=1}^{m_i} \exp \left[\sum_{h=1}^c \alpha_i (\theta_r - \delta_{ih}) \right]}, \quad (15)$$

where $p(x_{ik} | \theta_r, \alpha_i, \delta_{ik})$ denotes the probability of a person located at θ_r responding in item i ’s category k (i.e., x_{ik}) given item parameters α_i and δ_i ; for notational convenience, “ $\exp[\Xi]$ ” is used in lieu of “ e^Ξ .” As was the case with the 2PL model, the subscript on item i ’s discrimination parameter, α_i , indicates that items can differ in their discrimination. Additionally, δ_i represents item i ’s set of transition location parameters, δ_{ih} s, so that $\delta_i = [\delta_{i2}, \delta_{i3}, \dots, \delta_{im_i}]$. That is, the transition location parameter δ_{ih} reflects the transition from the $(h-1)$ response category to the (next) h^{th} category. Because Muraki arbitrarily defines the first transition location parameter as zero (i.e., $\delta_{i1} \equiv 0$), there are $m_i - 1$ transition locations (i.e., $\delta_{i2}, \delta_{i3}, \dots, \delta_{im_i}$); the number of response categories m_i is free to vary across items. (Note that we use italicized “ m ” to indicate the number of response categories and non-italicized “ m ” to indicate the number of transition locations [cf. MRFM].)

The probability of responding in a category as a function of the latent variable is graphically depicted by a response category’s *option response function* (ORF). Figure 8.4 contains an example ORF for a three-category item. Let us assume that our example item comes from a scale to measure quality of life and asks the respondent to rate the quality of his or her relationships on a three-category scale (1= “unsupportive/unsympathetic,” 2= “neutral,” and 3= “supportive/sympathetic”). As can be seen, a person who feels that he or she has a low quality of life (e.g., he/she is located at -2) would probably respond in category 1 as opposed to categories 2 or 3. In this case, applying the GPC model (Equation 15) yields a probability of 0.72 of responding in category 1. In fact, any person located below -1 has a higher probability of responding “unsupportive/unsympathetic” than in any of the other categories. This is what is represented by category 1’s ORF (the dashed line). Clearly, as the person’s quality of life increases, the greater the probability that a person will respond in category 2 and eventually in category 3. For example, a

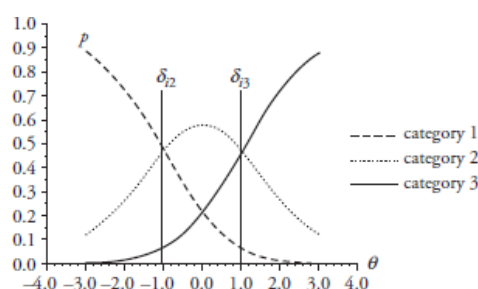
person with high quality of life (e.g., $\theta = 2.5$) has a probability of 0.81 of responding in category 3, a probability of 0.18 of responding in category 2, and 0.01 probability of responding in category 1. As can be seen from Figure 8.4, for a given θ , the sum of the probabilities across response categories is 1 (e.g., for $\theta = 2.5$ we have $0.81 + 0.18 + 0.01 = 1$).

Figure 8.4 also shows that the transition location parameter represents the intersection point of adjacent ORFs. With our three-category item ($m_i = 3$) we have two transition location parameters. Our first transition from response category 1 to category 2 occurs at δ_{i2} , and our second transition from category 2 to category 3 occurs at δ_{i3} . Although for our example item, the transition location parameters are in order, there is no requirement in the model that they be ordered. (An alternative to the GPC model is Samejima's [1969] graded response model.)

The GPC model can be simplified to obtain the Rasch *partial credit* model (Masters, 1982) by imposing the constraint that all items have the same discrimination. That is,

$$\Xi = \sum_{h=1}^k \alpha (\theta_r - \delta_{ih}). \quad (16)$$

Figure 8.4:



ORFs for an item with $\alpha_i = 1.0$, $\delta_{i2} = -1$ and $\delta_{i3} = 1$.

Note the omission of the item subscript on α . Like the GPC model, the PC model can be applied to ordered polytomous data as well as to dichotomous response data. In this latter case, the PC model simplifies to the Rasch/1PL model.

generalized rating scale and rating scale models

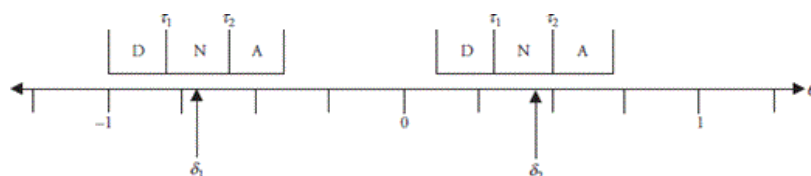
Although we can use either the GPC or PC models with rating scale data (e.g., a Likert response scale), if we believe or are willing to assume that the relative difficulty of endorsing of one rating over another is the same for all items using a common rating scale, then we can further simplify our models. In this case, we can decompose an item's transition locations, δ_{ih} s, into an item location parameter and a set of threshold components. That is, each item has a location on the latent continuum (δ_i), and the transition across adjacent rating categories is captured by a series of threshold parameters (τ_h s) that are constant for a common rating scale. As an example, imagine that we have two items that use a three-point rating format (D = Disagree, N = Neutral, and A = Agree). Figure 8.5 shows the item locations ($\delta_1 \cong -0.7$ and $\delta_2 \cong 0.4$) and the associated common set of threshold parameters (τ_1 and τ_2). Although items 1 and 2 are located at different points on the latent variable, the difficulty in endorsing the neutral category over the disagree category (represented by the threshold τ_1) is the same for both items. Similarly, the difficulty in endorsing the agree category over the neutral category (represented by the threshold τ_2) is also the same for both items. (Note that with four or more categories the thresholds do not have to be equidistant.) As can be seen, the thresholds are offsets from the item's location. Therefore, although the two items share a common set of thresholds, the actual location of the transition from—say, neutral to agree—occurs at different points on the latent variable's continuum. For example, for item 1, the location of the transition from neutral to agree occurs at approximately -0.65 , whereas for item 2, the transition occurs at $.5$.

The foregoing can generically be represented symbolically as $\delta_{ih} = \delta_i - \tau_h$, where δ_i is the item location and τ_h is the threshold between categories ($h - 1$) and h . Therefore, by substitution into Equation 14, the logistic deviate for the *generalized rating scale* (GRS) model (Muraki, 1990) is

$$\Xi = \sum_{h=1}^k \alpha_i (\theta_r - \delta_i + \tau_h). \quad (17)$$

p. 155 Muraki (1992) has interpreted the τ_h as the relative difficulty of “step” h “...in comparing other steps within an item” (p. 165); “difficulty” may also be interpreted as the difficulty of endorsing a particular category. Moreover, the τ_h s do not need to be sequentially ordered across the categories. The GRS allows items to vary in their capacity to discriminate among respondents located at different points on the latent continuum.

Figure 8.5:



Conceptualization of two items with one three-point rating scale.

The GRS model can be simplified to obtain the Rasch *rating scale* (RS) model (Andrich, 1978a, 1978b, 1978c) by imposing the constraint that all items have the same discrimination. That is,

$$\Xi = \sum_{h=1}^k \alpha (\theta_r - \delta_i + \tau_h). \quad (18)$$

(Note the omission of the item subscript on α to indicate a constant value of α across items.) As such, the RS model can be viewed as simplification of the GRS. Alternatively, one can view the RS model to be a constrained or reparameterized version (i.e., $\delta_{ih} = \delta_i - \tau_h$) of the PC model (Equation 16). In either case, the RS model simplifies to the Rasch/1PL model when applied to dichotomous data.

Nominal Response Model

The above polytomous IRT models assume that certain responses indicate more of what is being measured than do other responses. Thus, the corresponding responses categories contain information about the magnitude of the construct being measured by the item. In contrast, in some cases, the responses are *not* inherently ordered. In these cases, a response is simply distinct from the other responses. For example, assume that we are interested in measuring social anxiety. One of our items could be “I feel uncomfortable at parties” with a response format of “yes,” “no,” and “not applicable.” Because our format’s response categories cannot be ordered to reflect the degree of social anxiety any set of numbers that we use to represent the different response categories is arbitrary (e.g., 1= “yes,” 2= “no,” 3= “not applicable”; 1= “no,” 2= “not applicable,” 3= “yes”; etc.). In short, we are using a nominal (also known as a categorical) response format. As is the case with our ordered response formats, this nominal response format consists of mutually exclusive response categories.

To model the response behavior involving a nominal response format, we return to Equation 4 (for the reader’s convenience, it is presented here as Equation 19). This equation provides the log odds of a response of 1 (numerator) relative to a response of 0 (denominator). Stated another way, Equation 19 provides the log odds (logit) of a response in category 1 compared to a response in the baseline category 0:

$$\ln \left(\frac{p(x)}{1-p(x)} \right) = \Xi. \quad (19)$$

We can extend this idea to multiple response categories. That is, for each of our response categories, we can determine the log odds (or odds) that a respondent will select a particular response category relative to a baseline category. In terms of our social anxiety example, we might arbitrarily select the “not applicable” response as our baseline category and determine the log odds (or odds) of a respondent providing a “yes” or “no” response relative to a “not applicable” response. Moreover, rather than talking about the log odds (or odds) of a response in one category over another (baseline) category, we can directly express the probability of a particular response given the baseline category. Therefore, we modify Equation 5 to incorporate category parameters:

$$\Xi = \gamma_{ih} + \alpha_{ih} \theta_r, \quad (20)$$

where γ_{ih} and α_{ih} are, respectively, the h^{th} category intercept and slope parameters for item i 's logit regression line and θ_r is person r 's location on the latent construct. By substitution of Equation 20 into Equation 3 (and simplifying), we obtain Bock's (1972) *nominal response* (NR; also called the *nominal categories*) model:

$$p\left(x_{ik} \mid \theta_r, \alpha_i, \gamma_i\right) = \frac{\exp[\gamma_{ik} + \alpha_{ik}\theta_r]}{\sum_{h=1}^{m_i} \exp[\gamma_{ih} + \alpha_{ih}\theta_r]}, \quad (21)$$

where $p\left(x_{ik} \mid \theta_r, \alpha_i, \gamma_i\right)$ is the probability of responding in category k on item i , m_i is the number of response categories for item i , the vector α_i contains the m_i slope parameters ($\alpha_i = [\alpha_1, \dots, \alpha_{m_i}]$), and the vector γ_i contains the item i 's intercept parameters ($\gamma_i = [\gamma_1, \dots, \gamma_{m_i}]$). To identify the model, the baseline response category's slope and intercept are set to 0; by convention the baseline category is the response category with the highest frequency. Therefore, item i has $m_i - 1$ slope parameters and $m_i - 1$ intercept parameters. However, these $m_i - 1$ sets of parameters can be transformed so that each response category has a slope and intercept parameter subject to the constraints $\sum_{h=1}^{m_i} \alpha_{ih} = 0$ and $\sum_{h=1}^{m_i} \gamma_{ih} = 0$.

As is the case with the above polytomous models, the probability of responding in each response category as a function of the latent trait θ can be graphically depicted by the item's ORFs. In general, these have the appearance of those shown in Figure 8.4. That is, one ORF will be monotonically increasing (e.g., category 3 in Fig. 8.4), one will be monotonically decreasing (e.g., category 1 in Fig. 8.4), with any remaining ORFs appearing as unimodal and symmetric (e.g., category 2 in Fig. 8.4). To obtain the transition point (i.e., intersection) between the ORFs for categories k and k^* , $\delta_{k^*,k}$, we would calculate

$$\delta_{k^*,k} = \frac{\gamma_{k^*} - \gamma_k}{\alpha_k - \alpha_{k^*}}, \quad (22)$$

where $m_i > 2, k^* < k$, and $\alpha_{k^*} \neq \alpha_k$. When we have a dichotomous response format ($m_i = 2$), then the NR model reduces to the 2PL model. Moreover, by appropriately reparameterizing the NR model one can obtain the GPC and PC models. This hierarchical relationship would allow one to compare the relative fit of a model assuming a NR format with that of a model assuming an ordered format. Therefore, one has a way of investigating those situations in which we may believe responses should be ordered but are simply not sure on the order.

multidimensional two-parameter model

The foregoing models all include a single-person location parameter to denote a unidimensional construct. However, in some situations it is more plausible that multiple latent variables account for the observed data. For example, using our social anxiety example, we might hypothesize that there are two factors at the root of socially anxious behavior. One dimension is a self-consciousness factor with opposing endpoints of private and public self-consciousness, whereas the other dimension is a generalized anxiety factor. As such, these factors would be modeled using two latent variables: generalized anxiety and self-consciousness. We can model these data by extending the two-parameter model (e.g., Equation 5) to include F latent variables so that

$$\Xi = \gamma_i + a_i^\theta \eta_r, \quad (23)$$

where γ_i is the intercept of item i 's logit response plane, a_i^θ is a (row) vector containing item i 's discrimination parameters on the F latent variables (i.e., $a_i^\theta = [\alpha_{i1}, \dots, \alpha_{iF}]$), and η_r is a vector that contains person r 's location parameters on each of the F -dimensions ($\eta_r = [\theta_{r1}, \dots, \theta_{rF}]$). As is the case with the unidimensional 2PL model, γ_i is the interaction of the item's discrimination and location parameters, $\gamma_i = -\sum_{f=1}^F \alpha_{if} \delta_{if}$.

To obtain the probability of a response of 1 on item i given a person's latent locations, we substitute Equation 23 into our logistic distribution function, $\Psi(\Xi)$ (Equation 3) to obtain the *multidimensional compensatory 2PL* (M2PL) model:

$$p_i(x_i = 1 | \eta_r, a_i^\theta, \gamma_i) = \Psi\left(\gamma_i + a_i^\theta \eta_r\right) = \frac{e^{\gamma_i + a_i^\theta \eta_r}}{1 + e^{\gamma_i + a_i^\theta \eta_r}}. \quad (24)$$

Although we limit our presentation to the M2PL model for dichotomous data, it should be noted that there are multidimensional extensions of other dichotomous models as well as some polytomous models. In this latter case, however, at present there are no user-friendly packages to estimate the models' parameters. The M2PL model is an example of a multidimensional item response theory (MIRT) model.

Although the M2PL model is useful with multidimensional situations, it is sometimes convenient to have a single (i.e., scalar) value that represents the best that an item can discriminate across the latent variables. This value is known as the item's *multidimensional discrimination capacity*, A_i . Similarly, it is useful to have a single value that represents an item's "location" in the multidimensional space. Technically, item i 's *multidimensional item location*, Δ_i , is defined as the distance from the origin in the latent space (i.e., η) to the point of maximum discrimination in a particular *direction* from the origin. Further discussion of these concepts is beyond the scope of this chapter, and the reader is referred to Reckase (2009).

mixture item response theory model

An alternative multidimensional perspective can be found in a situation that involves a mixture of latent classes and latent continua. In this case, we conceptualize the latent variable as consisting of latent classes, within which are latent continua. For example, using our social anxiety example, we might hypothesize that social anxiety is best explained by a combination of *categorical* (mutually exclusive and jointly exhaustive) latent classes and a continuous latent variable rather than, as above, as two *continuous* factors. As such, the self-consciousness dimension is conceptualized as two discrete classes of homogeneous individuals. One of our classes consists of privately self-conscious persons, whereas the other latent class contains public self-consciousness individuals. Further, within each of these classes is a generalized anxiety continuum on which we can locate our respondents.

With this conceptualization, the observed data consist of one or more latent classes and within each latent class there is an IRT model. In the simplest case, there is only one latent class and the respondent sample

contains only members from this class and one has model–data fit with a simple IRT model; the respondent sample is also known as the *calibration sample*, and the process of obtaining estimates of person and item parameters is known as *calibration*. However, when the observed data consist of members from different latent classes, there is not an IRT model that accurately reflects the data for the entire calibration sample (i.e., there is model–data misfit). Rather, there are different item and person parameters that are conditional on the different subpopulations or latent classes. Mixture distribution models such as those of Rost (1990) as well as Mislevy and Verhelst (1990) have addressed this general idea, and their extensions of the Rasch model have been concerned with solution strategies that differ across subpopulations (for a general framework, see also Kelderman & Macready, 1990).

In the simplest case, our mixture model is

$$p_i(x_i = 1 | \theta_r, \alpha_\nu, \delta_{i\nu}, \nu) = \sum_{\nu} \pi_{\nu} \frac{e^{\alpha_{\nu}(\theta_r - \delta_{i\nu})}}{1 + e^{\alpha_{\nu}(\theta_r - \delta_{i\nu})}}, \quad (25)$$

where π_{ν} is latent class ν 's proportion and α_{ν} , and $\delta_{i\nu}$ are item i 's discrimination and location, respectively, in latent class ν . Just as is the case with the other IRT models, we are interested in obtaining item and person parameter estimates. However, with our mixture IRT model, our person parameters consist of not only a respondent's location on the latent continuum (θ) but also the individual's (latent) class membership. Therefore, each item has a location in each latent class's continuum. Similarly, each respondent has a location in each latent class's continuum and membership in only one class. This membership is probabilistic in nature. For example, person A has a probability of 0.8 of belonging to latent class 1 and a probability of 0.2 of belonging to latent class 2. Equation 25 may be extended to allow for varying item discrimination (i.e., $\alpha_{i\nu}$) as well as for applicability to polytomous data.

Estimation

Obtaining estimates for an instrument is referred to as calibrating the instrument. Generally speaking, some variant of maximum likelihood estimation (MLE) is the approach most commonly seen. The gist of MLE is to determine the parameter estimate that maximizes the likelihood function observed. To clarify what this means, let us, for the sake of simplicity and without loss of generality, assume dichotomous responses and a single latent variable, θ . Then the probability of a set of responses, x , on a L -item instrument is

$$p(x | \theta, \vartheta) = \prod_{i=1}^L (p_i^{x_i} (1 - p_i)^{(1-x_i)}), \quad (26)$$

where p_i is given by, say, the 1PL model; x_i is the response to item i ; and ϑ contains the item parameters. Once individual r 's responses are observed, this expression becomes a *likelihood function* and we have

$$L(x_r | \theta_r, \vartheta) = \prod_{i=1}^L (p_i^{x_{ri}} (1 - p_i)^{(1-x_{ri})}). \quad (27)$$

For computational reasons, we take the natural log (ln) of Equation 27 and obtain the *log-likelihood function*:

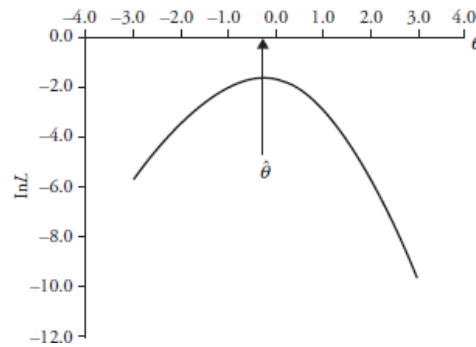
$$\ln L(x_r | \theta_r, \vartheta) = \sum_{i=1}^L x_{ri} \ln p_i + (1 - x_{ri}) \ln (1 - p_i). \quad (28)$$

The location of the maximum of the likelihood function is the same as that of the maximum of the log likelihood function.

p. 158 To help conceptually understand MLE, assume that we have a client, Kim, whose responses to a five-item depression scale are (1, 1, 0, 0, 0) where 1 = true and 0 = false—that is, Kim responded true to the first two items and false to the last three items. Moreover, assume that we are using the 2PL model for our depression scale and our item locations are $\delta_1 = -1, \delta_2 = -0.5, \delta_3 = 0, \delta_4 = 0.5$, and $\delta_5 = 1$ with item discriminations of $\alpha_1 = 1.5, \alpha_2 = 1, \alpha_3 = 1.25, \alpha_4 = 2$, and $\alpha_5 = 1.1$. In this example, we are interested in estimating Kim's location ($\hat{\theta}$) on our depression (latent) variable. In other words, we are searching for the value of Kim's location that most likely produces (i.e., maximizes the likelihood of observing) the responses of (1, 1, 0, 0, 0). The (log) likelihood function ($\ln L$) for Kim's response vector is shown in Figure 8.6. To find the value of Kim's location, one can envision traversing this function to find the location of its peak. This would be our estimate of Kim's depression. As can be seen, the function has its peak or maximum at approximately -0.3 . In other words, $\hat{\theta} = -0.3$ is the parameter estimate that maximizes the likelihood function for the observed responses of (1, 1, 0, 0, 0). (Below we return to how we can rescale this $\hat{\theta}$ to aid its interpretation.) The log likelihood function depicted in Figure 8.6 clearly has a nice parabolic shape and a single maximum value. In some circumstances, the (log) likelihood function does not have this shape. For example, if our observed responses consisted of a single value—say, all 1s or all 0s—then the (log) likelihood function would not have a maximum value. Graphically, the log likelihood function would increase and then become asymptotic to 0. Therefore, for certain response vectors that have zero variance, it is not possible to obtain a maximum likelihood estimate. However, there are alternative approaches that can be used. These Bayesian approaches still use the likelihood function but also utilize a prior distribution. The incorporation of the prior distribution with the likelihood function results in a distribution called the posterior distribution. The mode or mean of this posterior distribution is used as the estimate of the person parameter. When the mode is used, then the approach is known as maximum *a posteriori* (MAP), whereas when the mean is used the method is called expected *a posteriori* (EAP). To summarize, the primary ways we can estimate a person's location are by MLE, MAP, or EAP. (There are additional variations of MLE that can also be used.)

Although our example demonstrates the principles underlying the MLE of a person's location, the

Figure 8.6:



Log likelihood function for $x = (1, 1, 0, 0, 0)$.

same principle can be applied to estimating the item parameters. Further, variants of this basic MLE conceptualization, such as marginal MLE (MMLE), joint MLE (JMLE), or conditional MLE, address the additional complexities that arise when estimating item parameters in the context of unknown person parameters and vice versa, invoking prior distributions for item parameter estimation, as well as taking advantage of certain model properties. Greater details on these methods as well as other estimation approaches, such as Markov chain Monte Carlo (MCMC) or minimum chi-square, may be found in Baker and Kim (2004) and de Ayala (2009).

The nonlinearity of our IRT models requires the use of an iterative estimation approach that successively refines the parameter estimates until an acceptable level of refinement is attained. As such, model estimation is facilitated by using a computer program. Table 8.3 presents several estimation programs and the models that can be estimated with each of the programs. For example, for dichotomous models, such as the 1PL/Rasch, 2PL, and 3PL models, one could use BILOG-MG (Zimowski, Muraki, Mislevy, & Bock, 2003) program. With respect to polytomous models, the programs MULTILOG (Thissen, Chen, & Bock, 2003) and PARSCALE (Muraki & Bock, 2003) may be used; each of these programs can also estimate dichotomous models as well as unique models. The programs ConQuest (Wu, Adams, & Wilson, 1997), FACETS (Linacre, 1989, 2009), and WINSTEPS (Linacre, 2001) are Rasch model-focused programs (i.e., item discrimination is assumed to be one) and can be used with dichotomous and polytomous data; Conquest and FACETS can also be used to estimate the MFRM. Additionally, *Mplus* (Muthén, & Muthén, 2007), SAS (SAS, 2002), and SYSTAT (SYSTAT, 2007) may be used to estimate some of the models discussed. In contrast to the above, programs such as NOHARM (Fraser & McDonald, 2003), MINIFAC, BIGSTEPS, eRM, or ltm are available for free.

Table 8.3. Nonexhaustive List of Programs for Parameter Estimation

Program	Estimation method	Model (s) estimated
BILOG-MG	Items: MMLE	IP, 2P, 3P
	Persons: EAP, MAP, MLE	
ConQuest	Items: MMLE, JMLE	1PL/Rasch, PC, RS, LLTM, MFRM, MIRT
FACETS/MINIFAC ³	Items and Persons: JMLE	MFRM, 1PL/Rasch, PC, RS
NOHARM ³	Items: Ordinary LS ¹ on observed & predicted proportions	IP, 2P, 3P, MIRT models: 2P, M3P ⁴
Mplus	ML, Robust weighted LS ¹	IP/Rasch, 2P, GR, mixture IRT/Latent class mode
MULTILOG	Items: MMLE	IP, 2P, 3P, GR, PC, NR, MC
	Persons: EAP, MAP, MLE	
PARSCALE	Items: MMLE	IP, 2P, 3P, GR, GPC, GRS, PQRS, rater-effect
	Persons: EAP, MAP, MLE ²	
R (km add-on) (R, 2007) ³	Items: MMLE	1PL/Rasch, 2P, 3P, GR, GPC
R (eRm add-on) (R, 2007) ³	Items and Persons: Conditional MLE	1PL/Rasch, LLTM, PC, RS
SAS (IRT-FIT; NLMIXED)	Maximize an approximate integrated likelihood	IRT-FIT: 1PL/Rasch, 2PL, 3PL, GR, GPC, GRS, PC, RS, NR; NLMIXED: explanatory models
SYSTAT	Items: MLE	1PL, 2PL
	Person: MLE	
TESTFACT (Wood, Wilson, Gibbons, Schilling, Muraki, & Bock, 2003)	Full-information factor analysis	2P
WINMIRA (von Davier, 2001)		1PL/Rasch, PC, latent class analysis, mixture IRT model
WINSTEPS, BIGSTEPS ³ , MINISTEP ³	Items and Persons: JMLE	1PL/Rasch, PC, RS

Unless otherwise noted estimation can be performed on both normal ogive and logistic versions of models.

- 1 LS= Least-squares
- 2 Warm's Weighted MLE (WML)
- 3 Freeware
- 4 Modified 3P model—user provides pseudo-guessing parameter estimates

p. 160 Assumptions

All IRT models make assumptions about the nature of the data. Specifically, the IRT models discussed are predicated on a *dimensionality* assumption. This assumption states that the observations on the manifest variables are a function of one or more continuous latent person variables. Typically, this assumption is referred to as the *unidimensionality* assumption in the context of all models discussed above the M2PL model.

A second assumption is the *local or conditional independence* (CI) assumption. This assumption is the keystone of all estimation algorithms. The CI assumption states that for any group of individuals who are characterized by the same latent location(s), the conditional distributions of the item responses are all independent of each other (Lord & Novick, 1968). Therefore, whatever relationships exist among the items disappears when one conditions on the latent location(s).

The third assumption is the *functional form* assumption. This assumption states that the data follow the function specified by the model. For example, for Equation 8, the functional form states that (1) the probability of a response of 1 increases monotonically when there is an increase in θ , and (2) that for infinitely low θ s the probability of $x_i = 1$ approaches 0 (see Fig. 8.3).

The foregoing assumptions are common to all IRT models. Specific models make additional assumptions and/or demands of the data. For example, the 1PL model assumes that all items discriminate to the same degree, whereas the nominal response model assumes that respondents with infinitely low θ s will pick a particular option rather than randomly guessing at an item's options. Whether a model's unique assumptions are tenable needs to be examined in the context within which it is used.

Fit

To obtain the above mentioned advantages of IRT, it is necessary to have acceptable model-data fit. Implied in this statement is that we consider fit to be a matter of degree rather than absolute. As such, one question that needs to be asked is "What is our level of misfit tolerance?" Our simple answer is that, all things being equal, we will tolerate misfit up to the point where misfit interferes with our assessment objective. Moreover, it should be noted that model-data fit is necessary, but not sufficient, for obtaining validity evidence for an instrument.

Some of our model-data fit evidence will be obtained across all items (we call this instrument-level fit), whereas other evidence will be for each item (we call this item-level fit). Examples of instrument-level fit evidence include dimensionality assessment or model-level likelihood ratio statistics, whereas an example of item-level fit would be conditional dependence determination for a pair of items. In some cases, a lack of acceptable fit at the instrument-level may be explained by a lack of fit for a subset of items (i.e., if this

subset is removed, then one would have instrument-level fit). In other situations, it is possible to have evidence of instrument-level fit but not observe fit for each item. For example, we might have evidence supporting the use of a unidimensional model, but we also identify one item-pair exhibiting conditional dependence. (Whether this item-level misfit can be tolerated depends on the context.) As a result, we need to obtain evidence of fit at both the instrument and item levels.

Table 8.4 provides different aspects involved in assessing model-data fit along with example approaches. As can be seen, some aspects involve the tenability of assumptions because violations of assumptions may lead to inaccurate parameter estimates. Other aspects utilize some of the advantages of IRT over CTT discussed above (e.g., invariance) as vehicles to assess fit.

Table 8.4 shows that performing a fit analysis will generally involve using multiple programs. For example, to assess a model's dimensionality assumption would involve a statistical package that can perform a factor or principal component analysis or a specialized program (e.g., NOHARM for nonlinear factor analysis of binary data). However, because dimensionality assessment is only one aspect of a fit analysis, this step would then be followed by an examination of functional form, invariance, and conditional independence. As such, a proper fit analysis will involve not only an IRT calibration program but also a statistical package and/or a specialized program. Moreover, each estimation program provides its own approach to model-data fit analysis. For example, some programs will provide a Likelihood Ratio statistic (G^2) for instrument-level fit assessment, whereas others will also provide AIC and BIC statistics, and others will provide INFIT and OUTFIT statistics instead of G^2 , AIC, and BIC. This will also be true at the item level (i.e., the fit statistics in one program may not correspond to those available in another program).

p. 161 At the item level and in general, fit analysis approaches involve either fit statistics and/or ↴ graphical approaches. Typically, the fit statistics compare what is expected on the basis of the model with what was observed. In this regard, a nonsignificant fit statistic indicates a correspondence between what the model predicts and what is observed (i.e., fit). The graphical approach typically compares the predicted IRF (e.g., Fig. 8.3) to the observed IRF; with polytomous data the comparison would be between the predicted ORFs (e.g., Fig. 8.4) and the observed ORFs. When the predicted response function for an item corresponds with the observed response function, then there is evidence of fit. As such, item-level fit statistics and the graphical approach permit an evaluation of the functional form assumption. We believe that the graphical and statistical approaches are complementary and should always be used in conjunction with one another.

Table 8.4. Fit Analysis Aspects

Aspect	Approach example
Instrument-Level information	
	Assess dimensionality
	Dichotomous data: nonlinear factor analysis ^a (e.g., NOHARM)
	Polytomous data: linear factor analysis ^a or principal component analysis ^a
Overall model fit	Fit statistics: Likelihood Ratio Statistic (G^2), AIC, BIC, INFIT, OUTFIT
Item-Level information	
	Functional form
	Fit statistics (χ^2 , INFIT, OUTFIT)
	Graphical comparison of predicted with observed
	Dichotomous Data: predicted IRF and observed IRF
	Polytomous data: predicted ORF and observed ORF
Conditional independence	Q3 statistic (Yen, 1984), Residual Correlation
Invariance	Divide calibration sample into subgroups and compare the parameter estimates either graphically and/or statistically (we include differential item functioning within this aspect)

a Exploratory or confirmatory

Conditional item *dependence* may be observed when (1) using an IRT model with fewer latent traits than are necessary to correctly model the data (Yen, 1984; Tuerlinckx & De Boeck, 2001); (2) the response to one item increases the probability of a particular response to another item (i.e., item chaining or item interaction; Tuerlinckx & De Boeck, 2001; Yen, 1993); or (3) two or more items are related to one another because of some commonality, such as a group of reading comprehension items sharing a common passage. For an extensive list of additional situations in which conditional item dependence may occur see Yen (1993).

There are several ways to assess the tenability of the CI assumption. Two approaches are examining the residual matrix after fitting a factor model to data and Yen's Q_3 statistic (Yen, 1984, 1993). With the residual matrix approach we fit—say, a single-factor model (i.e., for one of our unidimensional models)—to the data and examine the residual matrix.⁵ If the values in the residual matrix are zero or very close to zero, then one has evidence of conditional item independence.

The Q_3 statistic is the correlation of the residuals (d_i, d_j) for an item-pair, i and j , after the person location estimates are partialled out. The residuals for items i and j are $d_i = x_i - p_i(\hat{\theta})$ and $d_j = x_j - p_j(\hat{\theta})$, respectively. The terms $p_i(\hat{\theta})$ and $p_j(\hat{\theta})$ are the probability of a correct response on items i and j , respectively, according to an IRT model using the estimated item parameters and the location estimate ($\hat{\theta}$). Because the item responses used in calculating the correlations are also used in estimating the person's location, Q_3 is expected to be negatively biased (Yen, 1984). When conditional independence holds, then the expected value of Q_3 is approximately $-1/(N - 1)$, where N is the sample size (Yen, 1993). Critical values for flagging the existence of CID with Q_3 do not exist. Therefore, in practice, a cut-point for Q_3 of ± 0.2 has been used for identifying items that are exhibiting conditional dependence. Alternatively, one can conduct a simulation study in which conditional independence is true and obtain the optimal cut-points for Q_3 for a given situation (e.g., a set of item parameter estimates, sample size, etc.).

The invariance aspect of model-data fit assessment capitalizes on one of the IRT advantages over CTT. As mentioned above, item- (and person-) parameter estimates are invariant when one has model-data fit. Thus, if we observe invariance in our item parameter estimates, then we have evidence supporting model-data fit. By "invariance of parameter estimates" we mean that our estimates are the same up to a linear transformation. As an example of invariance, assume that we have five items with which we calibrate, for simplicity and without loss of generality, the 1PL model; assume that we have model-data fit. Our calibration produces item location estimates ($\hat{\delta}_i$): item 1 ($\hat{\delta}_1 =$) of -2 , item 2 ($\hat{\delta}_2 =$) of -1.75 , item 3 ($\hat{\delta}_3 =$) of -1.5 , item 4 ($\hat{\delta}_4 =$) of -1.25 , and item 5 ($\hat{\delta}_5 =$) of -1 . Another administration of our five items to a different sample produces the estimates of $\hat{\delta}_1 = -1$, $\hat{\delta}_2 = -0.75$, $\hat{\delta}_3 = -0.5$, $\hat{\delta}_4 = -0.25$, and $\hat{\delta}_5 = 0$; again assume that we have model-data fit. As can be seen, our second set of estimates is simply shifted up the continuum by one logit. I can transform the second set of estimates to be on the same metric as the first by simply subtracting 1 (i.e., estimate $_{NEW} = \text{estimate}_{OLD} - 1$) from each estimate. Conversely, I can transform the first set of estimates to be on the same metric as the second by adding 1. Moreover, the correlation between the two sets of estimates is perfect with or without the transforming one metric to be the same as the other.

The gist of invariance assessment is to divide the calibration sample into two subsamples. The two subsamples can be created by random assignment and/or on the basis of a particular interest (e.g., males and females, high person locations and low person locations, etc.). Each of the subsamples is separately calibrated and the item parameter estimates compared to one another. There are several ways of making these comparisons, such as the correlation of item parameters, calculating the Mantel-Haenszel statistic, and/or the calculation of the root mean square difference (RMSD) between IRFs (or ORFs).

For the correlational approach, we simply calculate the correlation for a given parameter across our subsamples. For example, for two subsamples, S and T , and the 2PL model we would have two correlations, one for item discrimination and one for item location. Thus, we would calculate the Pearson product-moment correlation between subsample S and subsample T 's discrimination estimates as well as between the two subsamples' sets of item location estimates. A large value for the correlation—say, above 0.9—would provide evidence of invariance across our subsamples.

The correlational approach is sufficient for models that contain only a single-item parameter (e.g., the 1PL model). However, with multi-item parameter models (e.g., the 2PL model) the correlation does not reflect the interaction of the item's parameters represented in the item's IRF (or ORF). Therefore, to simultaneously compare the item parameter estimates across subsamples, one needs to, in effect, compare an item's IRFs (or ORFs) across subsamples. The *RMSD* can be used for making this comparison.

To calculate an item's *RMSD* we need to specify a range of interest. Typically, this range is from -3 to 3 . We then subdivide this range into W equally spaced θ s. For example, if our range is -3 to 3 we can divide it

into 121 equally spaced θ s using logit increments of 0.05 (i.e., $-3, -2.95, -2.9, \dots, 3$); the smaller the increment the greater the index's accuracy as a measure of the difference between the two IRFs. Then, the $RMSD$ for item i is given by

$$RMSD_i = \sqrt{\frac{\sum_{w=1}^W [p_{iS}(\theta_w) - p_{iT}(\theta_w)]^2}{W}}, \quad (29)$$

where $p_{iS}(\theta_w)$ and $p_{iT}(\theta_w)$ are calculated using the item parameter estimates from subsamples S and T , respectively; θ_w is the w^{th} θ value in the range of interest (e.g., $-3, -2.95, -2.9, \dots, 3$); and W is the number of equally spaced θ s in the range of interest (e.g., $W = 121$). Conceptually, $RMSD_i$ is the average absolute distance between the two IRFs. When $RMSD_i$ equals 0, then there is no difference between the two IRFs. However, one should expect that even with perfect model-data fit that estimation error will be reflected in an item's non-zero, albeit small, $RMSD_i$ value. From this perspective, a small $RMSD_i$ reflects two IRFs that may be considered to be sufficiently similarly to not be reason for concern (i.e., subsample S 's IRF would fall within the confidence band for subsample T 's IRF).

p. 163 In those cases where one observes a large $RMSD_i$, there may be various reasons for its magnitude. For example, the item may be poorly written and thereby interpreted differently across the subsamples or the model may have insufficient item parameters to accurately describe the item. Depending on the diagnosis of the cause(s) of the magnitude of $RMSD_i$, one may decide to omit the item from the instrument and retain only those items with small $RMSD_i$ values.

Unlike the correlational approach that yields a correlation across items, with $RMSD_i$ we have one value for each item. One approach to obtaining an instrument-level invariance assessment is to calculate the difference between the subsamples' respective *total characteristic functions* (TCFs). The TCF is based on the expected trait score, $\mathcal{ET}$ (T is the Greek letter tau):

$$\mathcal{ET} = \sum_{i=1}^L p_i(\theta), \quad (30)$$

where θ can be a value from a range of interest or a person's estimated location, L is the instrument's length, and p_i is given by one of our dichotomous models. (The expected trait scores for polytomous models may be found in de Ayala [2009].) Combining Equation 30 with the idea symbolized by Equation 29, we have that our instrument-level invariance assessment, $RMSD_{TCF}$, is given by

$$RMSD_{TCF} = \sqrt{\frac{\sum_{w=1}^W \left[\left(\sum_{i=1}^L p_{iS}(\theta_w) \right) - \left(\sum_{i=1}^L p_{iT}(\theta_w) \right) \right]^2}{W}}, \quad (31)$$

where θ_w and W are defined as above, the first term in the numerator is the expected trait score for subsample S (i.e., $\mathcal{E}T_S = \sum p_{is}(\theta_w)$), and the second term is the expected trait score for subsample T (i.e., $\mathcal{E}T_T = \sum p_{iT}(\theta_w)$). A value of $RMSD_{TCF}$ close or equal to zero would provide evidence of invariance. Both $RMSD_{TCF}$ and $RMSD_i$ should be used in conjunction with plots of the TCFs and IRFs to determine whether the magnitude of these statistics is representative of a systematic difference across the continuum or reflects a difference for a particular portion of the continuum.

Although $RMSD_{TCF}$ and $RMSD_i$ allow us to simultaneously compare an item's parameters across subsamples, there is a price to pay for this convenience. Specifically, prior to their use, we need to align the subsamples' metrics to one another, otherwise the magnitude of $RMSD_{TCF}$ and $RMSD_i$ may reflect, in part, the differences in the two metrics. The alignment of metrics (also known as *linking*) is discussed in the [next section](#).

As mentioned in Table 8.4, we include *differential item functioning* (DIF) in the invariance aspect. Differential item functioning is defined as an item that displays different statistical properties for different manifest groups after the groups have been matched on a proficiency measure (Angoff, 1993). In the DIF nomenclature, one of the manifest groups is known as the *focal group*, whereas the other is called the *reference group*. The focal group (e.g., females) is the one being investigated to see if it is disadvantaged by the item. The reference group is the comparison group (e.g., males). Graphically, DIF can be represented as the difference between two IRFs: one IRF is based on the item's parameter estimate(s) from the focal group and the other IRF is based on the item's parameter estimate(s) from the reference group. If an item is not exhibiting DIF, then the groups' IRFs would be superimposed on one another (i.e., within sampling error) after we link the two groups' metrics. However, if the item is exhibiting DIF, then the two IRFs are not superimposed after we link the two groups' metrics. Therefore, the existence of DIF means that the DIF item's parameter estimates are not invariant across the manifest groups (i.e., item-level misfit).

Although defined in terms of proficiency assessment, DIF is potentially applicable to nonproficiency assessments. As an example, we return to our social anxiety example. As part of our fit analysis, we perform separate calibrations for males and females. If we find that females respond differently to an item than males, even after we account for their respective locations on the social anxiety continuum, then our item is exhibiting DIF. It may be that the item's text elicits a different interpretation by female respondents than in male respondents (e.g., the text is sexist).

There are a number of approaches for assessing DIF. Two of these approaches are the *Mantel Haenszel* statistic (MH) and the use of *logistic regression* (LR). The MH statistic allows us to determine whether the responses to an item are independent of group membership after conditioning on the observed scores; MH is evaluated against the standard χ^2 critical values with degrees of freedom equal to 1. LR is a technique for making predictions about a binary variable from one or more quantitative and/or qualitative variables. In the current context, the binary variable is the response to an item and the predictors might be gender and/or some measure of the construct; Zumbo (1999) as well as French and Miller, (1996) discuss the technique's application to ordinal responses. As such, we logistically regress the responses to an item on the construct measure and/or on a manifest group indicator (e.g., gender). Conceptually, the application of LR to DIF analysis requires performing a logistic regression analysis for an item using members of the reference group and a second analysis for the same item with members of the focal group. The group results are compared using the ΔG^2 statistic. For both the MH and LR approaches, a nonsignificant test statistics indicate that DIF was not detected. See Camilli and Shepard (1994) for more information on DIF analyses.

Metric Transformations and Linking

Examination of our models shows that there is an indeterminacy of our parameter estimates. As an example, consider the 2PL model (Equation 8). We can add or subtract a constant from θ and δ_i and not change the logistic deviate. As a result, the IRF is unaffected although its location moves up or down the continuum. Stated another way, the origin of the metric is arbitrary. Similarly, multiplying θ and δ_i by a constant and dividing α_i by the same constant would leave $\alpha_i (\theta_r - \delta_i)$ unchanged. This implies that the unit for measuring θ and δ_i is also arbitrary. This indeterminacy is addressed in different ways by different programs. Thus, the program's user does not have to be concerned about this matter *per se*. However, we mention it because this issue facilitates the transformation of our metric to have certain characteristics that facilitate interpretation of the scale or to align two metrics with one another. The need to align metrics would occur if we administer an instrument to two samples or administer alternate forms of an instrument to a sample or to different samples.

We can rescale our parameters or their estimates by using the metric *transformation coefficients* ζ and κ . The values of ζ and κ may be given for a particular scale, such as the T-score scale (i.e., $\zeta = 10, \kappa = 50$), or they may be calculated to transform one metric to be the same as another metric (e.g., for use with calculating $RMSD_{TCF}$ and $RMSD_i$). One simple approach for calculating ζ and κ uses the means and standard deviations of the item locations. In this approach, the transformation coefficient ζ is obtained by taking the ratio of the two metrics' item location standard deviations

$$\zeta = \frac{s_{\delta^*}}{s_{\delta}}, \quad (32)$$

where s_{δ^*} is the standard deviation of the item locations on the *target metric* and s_{δ} is the standard deviation of the of the item locations on the *initial metric*. (The initial metric is the metric that is transformed to align with the target metric.) Once ζ is determined, the other transformation coefficient κ is obtained by

$$\kappa = \bar{\delta}^* - \zeta \bar{\delta}, \quad (33)$$

where $\bar{\delta}^*$ is the mean of the item locations on the target metric and $\bar{\delta}$ is the mean of the item locations on the initial metric.

An alternative approach for determining ζ and κ is known as the total (or test) characteristic curve method (Stocking & Lord, 1983). In this method, the total characteristic functions are obtained for the initial and target metrics. The values of ζ and κ that align the TCF on the initial metric with the TCF on the target metric are determined by minimizing the differences between the two TCFs. Typically, this is done by using a program such as EQUATE (Baker, 1993b), ST (Hanson & Zeng, 2004), or POLYST (Kim & Kolen, 2003).

Once the values of ζ and κ are known, then we can use them to transform our item and person parameters. Each item's discrimination parameter (or its estimate) is transformed by

$$\alpha_i^* = \frac{\alpha_i}{\zeta}, \quad (34)$$

where α_i is item i 's discrimination on the initial metric and α_i^* is the transformed discrimination value on the target metric. In terms of a slope–intercept parameterization, the item–wise transformation would be

$$\gamma_i^* = \gamma_i - \frac{\alpha_i \kappa}{\zeta}. \quad (35)$$

To transform our item locations, we use the standard linear transformation of

$$\xi^* = \zeta(\xi) + \kappa, \quad (36)$$

where ξ represents the parameter on the initial metric to be transformed (e.g., δ_i or its estimate) and ξ^* represents the same parameter on the target metric. For example, to transform our item locations, δ_i , to be on another metric we would use $\delta_i^* = \zeta(\delta_i) + \kappa$. (Sometimes the application of Equation 36 with $\xi = \theta$ and $\xi^* = \theta^*$ is known as *equating*.)

p. 165

Equations 32 through 36 are used to transform one (initial) metric to another (target) metric so that we can subsequently use our estimates interchangeably across samples and/or alternate forms. Another transformation that is sometimes useful is the conversion of our θ scale to one that may be intrinsically useful. As mentioned above, we can apply Equation 36 with $\xi = \theta$ to transform our θ scale to, for example, a T-score scale. Alternatively, we may to transform our θ scale to be on an observed or summed score scale. For example, assume that we have a 10-item social anxiety scale that uses a true/false response format. When we convey a person's location on the social anxiety continuum, it may be more useful to the respondent to know that he or she has a score of 4 on an 11-point scale (i.e., 0, 1, ..., 10) rather than a -1 (on an infinite θ scale). We can perform this transformation by using the expected trait score, ET (Equation 30), with $L = 10$ and θ (actually our estimate of) equal to -1 , and calculating each item's probability according to one of our dichotomous model (e.g., the 2PL model). The sum of these 10 probabilities would be our expected trait score for a person located at -1 .

Although our expected trait score is on the summed score scale, it is simply a conversion of the IRT metric and only has the *appearance* of a summed score. Therefore, we still have all benefits and advantages of IRT (e.g., neither θ nor $\mathcal{ET}$ depend on the distribution of persons, invariance, etc.). Moreover, because for a given calibration each θ will yield the same $\mathcal{ET}$, we can create a concordance table (i.e., a table that shows for each θ what the corresponding $\mathcal{ET}$ is) or graph the total characteristic curve to facilitate the conversion of a θ to its $\mathcal{ET}$; this curve has an ogival pattern (e.g., Fig. 8.3) with $\mathcal{ET}$ on the ordinate.

Calibration Sample Size

It cannot be stressed enough that sample size guidelines should not be interpreted as hard and fast rules. Specific situations may require more or fewer persons than other situations given the (mis)match between the instrument's range of item locations and the sample's range of person locations, response data characteristics (e.g., missing data, for polytomous data the distribution of responses across categories), and the purpose of the instrument's administration (e.g., establishing norms).

Some factors that also need to be considered in determining sample size are the calibration model, the estimation procedure and its possible interaction with instrument characteristics (e.g., instrument length), the desired degree of estimation accuracy of items and/or persons, model-data fit diagnostics (e.g., ancillary technique sample size requirements, fit statistics' power), and the use of prior distribution(s); in a case using a prior distribution, the match with the population distribution also requires consideration.

As an example of the interaction of some of these factors, consider JMLE and MMLE. With JMLE our instruments should consist of at least 20 to 25 items to minimize estimation bias, whereas with MMLE the length is not important for estimation (all things being equal). However, the instrument's length does have implications for the veracity of chi-squared fit statistics. That is, with MMLE we could calibrate an instrument that had less than 20 items, but we should not use the item-level chi-squared fit statistics as part of our model-fit analysis. Additionally, with small calibration samples, our plots of predicted and observed IRFs (or ORFs) would be less useful and the power of our fit statistics would be adversely affected.

Although some have suggested that "useful information can be obtained from samples as small as 100" (Wright, 1977, p. 224), typically sample sizes are substantially larger. (Wright's statement was with respect to the Rasch model, and it was prefaced by a statement of a desirable sample size of 500 or more.) Another caveat about smaller calibration samples (i.e., 100 or less) is that with a smaller sample size there is a higher probability, all other things being equal, that everyone will provide the same response (e.g., a response of 0) to one or more items. In these cases one cannot estimate the item parameter(s). The same problem may occur with "short" instruments. That is, with short instruments there is an increased chance of a person providing the same response for all items. As such, this individual's location could not be estimated using MLE.

Assuming MMLE, the use of a prior distribution for estimating item discrimination and favorable conditions (e.g., θ /prior distribution match, etc.), it appears that a calibration sample size of at least 500 persons tends to produce reasonably accurate item parameter estimates. If we also assume that the respondents distribute themselves across the response categories in reasonable numbers, then this guideline would also be applicable for the GPC and GRS models. With the NR model and making the same assumptions as above, then we suggest that the minimum sample size be 600. Of course, having more respondents than these minima is preferable. However, for all our models we anticipate that there is a sample size—for example, 1200 to 1500 or so—at which one reaches, practically speaking, a point of diminishing returns in terms of improvement in estimation accuracy for a given model. (These maxima should not be interpreted as upper bounds.) It must be noted that less favorable situations may necessitate larger sample sizes. To reiterate, the caveats and considerations previously mentioned as well as not interpreting sample size guidelines as hard and fast rules still apply to our recommendation.

With models that do not require estimation of item discrimination and given the foregoing caveats and considerations at the beginning of this section, then a rough sample size guideline is that a calibration sample should have at least a couple hundred respondents. This should not be interpreted as a minimum but, rather, as a reasonable compromise. Certain applications may require more respondents, whereas in others a smaller sample may suffice.

Summary

Item response theory is a latent variable modeling approach that is focused on item responses and their relationship to one or more latent variables that are our constructs of interest. We may apply IRT in two-facet situations (i.e., person by items) or in cases that have more than two-facet cases, such as to person by items by judges data (i.e., three-facets). Item response theory models may be used to fulfill either a descriptive or explanatory (predictive) objective. Further, IRT models may be used with discrete dichotomous and polytomous response data to obtain parameter estimates for items and people that are on a continuous scale.

In contrast to Classical Test Theory, IRT offers a number of benefits and advantages, such as person and item parameter invariance, and the capacity to design instruments that have specific psychometric properties, such as equiprecise measurement. However, to realize these benefits and advantages requires having model-data fit. The assessment of model-data fit involves determining the tenability of the model's assumptions as well as the presence of IRT properties (e.g., invariance).

Obtaining the item and person parameter estimates is typically accomplished via MLE (conditional or joint) or by MMLE. With MMLE the item parameters are estimated separately from estimating the person parameters. Because this separation allows one to first determine whether there is model-data fit for the instrument before estimating persons, MMLE is a more computationally efficient approach than JMLE. Estimating person parameters can be accomplished using MLE or by one of the Bayesian approaches of expected *a posteriori* and maximum *a posteriori*. These Bayesian approaches allow estimating a person's location in cases where MLE fails.

Once we have estimated our parameters, we have obtained a metric for our latent continuum. This is a relative, not an absolute, metric. As a result, our estimates can be transformed to (1) align the metrics from different samples and/or alternate forms of our instrument, (2) facilitate interpretations, and/or (3) to make comparisons (e.g., across groups, instruments, and/or longitudinally). It should be noted that the successful application of IRT does not preclude the necessity of obtaining validity evidence for an instrument.

Future Directions

In the following, I present some areas that either are or should see greater interest as well as areas in need of greater research. To expand the use of IRT in other fields (e.g., Industrial-Organizational, clinical field), future research needs to develop estimation procedures for small-sample calibration. At present, IRT is constrained to large samples, and this presents an impediment to applying it to situations with 30 or 50 respondents. In this regard, the use of a Bayesian perspective may be beneficial for obtaining estimates.

Separate from small sample estimation, but still within a Bayesian framework, we have our second area of interest—Markov chain Monte Carlo (MCMC) techniques for estimation. Markov chain Monte Carlo primary advantages are its flexibility and adaptability. In this regard, it allows one to experiment with new models “relatively” easily because one does not have to develop and validate complicated estimation algorithms to estimate these new (complicated and/or intricate) models. Moreover, MCMC would be particularly attractive when these models are of specific interest in a given context, and as a result, one could not justify a large time investment with limited utility. (It should be noted that MCMC's proper use requires a sophisticated understanding of the particular MCMC algorithm being used.) Another area within the Bayesian perspective is the use and advancement of posterior predictive checking for examining IRT model-data fit.

The above IRT models are applicable to single-level data collection schemes. However, because these models are examples of the generalized linear model, it is possible to extend our models to multilevel data

p. 167 collection. These multilevel situations (also known as hierarchical) arise in, for example, the three-facet reference frames (e.g., judges rating people's responses). Although these can be modeled using the MFRM, if we use a multilevel IRT model, then we do not have to assume conditional independence among our judges. Other multilevel cases would involve items that are chained together or that share a common passage, cross-national assessments, longitudinal data, and so on. In short, multilevel IRT would have applicability with clustered/nested data and assessment contexts in which conditional dependence is present. In the latter, one could also use a polytomous model or a testlet model (Wainer, Bradlow, & Wang, 2007) as an alternative to a multilevel IRT model.

The last areas that we present are cognitive diagnostic modeling and automated item generation. In cognitive diagnostic modeling, we attempt to classify individuals in terms of his or her mastery of skills. These skills can be those that are used to correctly answer items in a proficiency assessment situation or coping skills for dealing with, for example, social anxiety. By automated item generation, we mean the creation of items in real time and as needed according to a cognitive model or according to a cognitive diagnostic model. This marriage would allow for a dynamic assessment process, more completely incorporate cognitive psychology into assessment, and more fully exploit the computerized administration platform. Moreover, by incorporating cognitive diagnostic capabilities into the item generation, we would have an automated system that would provide each respondent with a profile of information, such as his or her location(s), which skills are mastered, and which skills need to be mastered. Depending on how the system is implemented some of the above models (e.g., LLTM, 2PL, M2PL, mixture models) or Adams and Wilson's random coefficient multinomial logit model (or its multidimensional variant [Adams, Wilson, & Wang, 1997]) are applicable.

Author Note

RJ de Ayala, Quantitative, Qualitative, and Psychometric Methods Department of Educational Psychology Teachers College Hall 114 University of Nebraska - Lincoln, Lincoln, NE 68588, USA.

Notes

1. Probabilities may be expressed in terms of the odds of an event occurring and vice versa. The probability of event x occurring expressed in terms of the odds of x is

$$p(x) = \frac{\text{odds}(x)}{1 + \text{odds}(x)}. \quad (37)$$

Expressing the odds in terms of probabilities gives us

$$\text{odds}(x) = \frac{p(x)}{1 - p(x)}. \quad (38)$$

Substituting of Equation 3 for $p(x)$ in Equation 38 and simplifying leads to

$$\text{odds}(x) = e^{\Xi}. \quad (39)$$

Because odds have a range from 0 to ∞ , with a value of 1 reflecting no difference between the event occurring and not occurring, we have an asymmetry in the odds scale. As a result, the odds of an event are sometimes transformed to the (natural) logarithmic scale (i.e., $\ln[\text{odds}(x)]$). On the log scale, a value of 0 reflects no difference between the event occurring and not occurring, positive values indicate that the odds of success (e.g., $x = 1$) are greater than of failure, and negative values reflect that the odds of failure (e.g., $x = 0$) are greater than for success. This transformation gives the *log odds* or the *logit* of the event occurring. Therefore, taking the natural log of both sides of Equation 39 gives

$$\ln[\text{odds}(x)] = \ln\left[\frac{p(x)}{1 - p(x)}\right] = \Xi. \quad (40)$$

2. In a proficiency assessment context, individuals at the lower end of the latent continuum may be expected to have a non-zero probability of providing a response of 0. For example, examinees that have low mathematics proficiency may be expected to incorrectly respond to, say, a topology question on a mathematics examination. If this mathematics examination uses a multiple-choice item format, then some of these low-proficiency individuals may select the correct option simply by guessing. In these cases the item's response function has a lower asymptote that is not asymptotic with 0 but with some non-zero value. The *three-parameter* model addresses this non-zero lower asymptote. The three-parameter model can be viewed as an extension of the two-parameter model. To explain this we need to be concerned with two cases. The first case is "What is the probability of a response of 1 on an item when an individual responds consistent with his or her location on θ ?" In this case, our answer is that the probability of the response of 1 is modeled by the 2PL model. The second case to consider is "What should be the probability of a response of 1 on an item due to chance alone?" To answer this question, let us symbolize this probability as χ_i . Therefore, when a person can be successful on item i on the basis of chance alone (i.e., irrespective of the person's location), then the corresponding probability is given by $\chi_i(1 - p_i)$. In this case, as θ becomes progressively more negative, then p_i approaches 0 and $\chi_i(1 - p_i)$ simplifies to χ_i . Stated another way, the probability of a response of 1 for an individual with an infinitely low location is χ_i . As such, χ_i represents the IRF's lower bound or asymptote. Putting these two (mutually exclusive) cases together, we can obtain the probability of a response of 1 from

$$p_i^* = p_i + \chi_i (1 - p_i), \quad (41)$$

where p_i is given by the two-parameter model. Equation 41 may be rearranged to be

$$p_i^* = \chi_i + (1 - \chi_i) p_i. \quad (42)$$

By substitution of the 2PL model for p_i in Equation 42, we obtain the *three-parameter logistic* (3PL) model

$$p_i^*(x_i = 1 | \theta_r, \alpha_i, \delta_i, \chi_i) = \chi_i + (1 - \chi_i) \frac{e^{\alpha_i(\theta_r - \delta_i)}}{1 + e^{\alpha_i(\theta_r - \delta_i)}}, \quad (43)$$

where χ_i is item i 's *pseudo-guessing* or *pseudo-chance* parameter and is the probability of a response of 1 when θ approaches $-\infty$; δ_i and α_i are defined as above. Therefore, with the 3PL model, there are three parameters characterizing item i (i.e., $\alpha_i, \delta_i, \chi_i$). Because there is a normal ogive version of the three-parameter model, Equation 35.43 is sometimes presented incorporating the scaling factor D .

Bibliography

Adams, R.J., Wilson, M., & Wang, W. (1997). The multidimensional random coefficients multinomial logit model. *Applied Psychological Measurement*, 21, 1–23.

[Google Scholar](#) [WorldCat](#)

Andrich, D. (1978a). Relationships between the Thurstone and Rasch approaches to item scaling. *Applied Psychological Measurement*, 2, 449–460.

[Google Scholar](#) [WorldCat](#)

Andrich, D. (1978b). A rating formulation for ordered response categories. *Psychometrika*, 43, 561–573.

[Google Scholar](#) [WorldCat](#)

Andrich, D. (1978c). Application of a psychometric rating model to ordered categories which are scored with successive integers. *Applied Psychological Measurement*, 2, 581–594.

[Google Scholar](#) [WorldCat](#)

Angoff, W.H. (1993). Perspectives on differential item functioning methodology. In P.W. Holland & H. Wainer (Eds.), *Differential Item Functioning* (pp. 3–23). Hillsdale, NJ: Lawrence Erlbaum.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Assessment Systems Corporation. (1997). *XCALBRE*. St. Paul, MN: Author.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Baker, F.B. (1993a). Sensitivity of the linear logistic test model to misspecification of the weight matrix. *Applied Psychological Measurement*, 17, 201–210.

[Google Scholar](#) [WorldCat](#)

Baker, F.B. (1993b). EQUATE 2.0: A computer program for the characteristic curve method of IRT equating. *Applied Psychological Measurement*, 17, 20.

[Google Scholar](#) [WorldCat](#)

Baker, F.B., & Kim, S. -H. (2004). *Item response theory: Parameter estimation techniques* (2nd edition). New York: Marcel Dekker.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Bock, R.D. (1972). Estimating item parameters and latent ability when responses are scored in two or more nominal categories. *Psychometrika*, 37, 29–51.

[Google Scholar](#) [WorldCat](#)

Camilli, G. & Shepard, L.A. (1994). *Methods for identifying biased test items*. Thousand Oaks, CA: Sage.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

de Ayala, R.J. (2009). *The theory and practice of item response theory*. New York: The Guilford Press.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Drasgow, F., & Olson-Buchanan, J.B. (Eds.) (1999). *Innovations in Computerized Assessment*. Hillsdale, NJ: Lawrence Erlbaum.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Embretson, S.E. (1984). A general latent trait model for response processes. *Psychometrika*, 49, 175–186.

[Google Scholar](#) [WorldCat](#)

Embretson, S.E. (1985). *Test Design*. New York: Academic Press.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Embretson, S.E. (1996). Cognitive design principles and the successful performer: A study on spatial ability. *Journal of Educational Measurement*, 33, 29–40.

[Google Scholar](#) [WorldCat](#)

Fischer, G. (1973). The linear logistic test model as an instrument in educational research. *Acta Psychologica*, 37, 359–374.

[Google Scholar](#) [WorldCat](#)

Fraser, C., & McDonald, R.P. (2003). *NOHARM: A Windows program for fitting both unidimensional and multidimensional normal ogive models of latent trait theory* [Computer Program]. Niagara College, Welland, Ontario, Canada. Available at people.niagaracollege.ca/cfraser/download. Last accessed July 16, 2012.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Frederiksen, N., Mislevy, R.J., & Bejar, I.I. (1993). *Test Theory for a New Generation of Tests*. Hillsdale, NJ: Lawrence Erlbaum.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

French, A.W., & Miller, T.R. (1996). Logistic regression and its use in detecting differential item functioning in polytomous items. *Journal of Educational Measurement*, 33, 315–332.

[Google Scholar](#) [WorldCat](#)

Haertel, E.H. (2006). Reliability. In R. Brennan (Ed.), *Educational Measurement*, (4th ed., pp. 65–110). Westport, CT: American Council on Education/Praeger.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Hanson, B., & Zeng, L. (2004). *ST: A computer program for IRT scale transformation* [Computer Program]. Iowa City, IA: ACT. Program available at www.education.uiowa.edu/centers/casma/computer-programs.aspx. Last accessed July 16, 2012.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Irvine, S.H., & Kyllonen, P.C. (2002). *Item Generation for Test Development*. Hillsdale, NJ: Lawrence Erlbaum.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Kelderman, H., & Macready, G.B. (1990). The use of loglinear models for assessing differential item functioning across manifest and latent examinee groups. *Journal of Educational Measurement*, 27, 307–328.

[Google Scholar](#) [WorldCat](#)

Kim, S., & Kolen, M.J. (2003). *POLYST: A computer program for polytomous IRT scale transformation* [Computer software]. Iowa City, IA: University of Iowa. Program available at www.education.uiowa.edu/centers/casma/computer-programs.aspx. Last accessed July 16, 2012.

Linacre, J.M. (1988). *Facets: A computer program for many-facet Rasch measurement*. Chicago IL: Winsteps.com.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Linacre, J.M. (1989). *Many-facet Rasch measurement*. Chicago IL: MESA Press.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Linacre, J.M. (2001). *A user's guide to WINSTEPS/MINISTEPS*. Chicago IL: Winsteps.com.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Linacre, J.M. (2009). *A user's guide to FACETS: Rasch-Model computer programs*. Chicago IL: Winsteps.com.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Lord, F.M. (1952). A theory of test scores. *Psychometric Monograph*, No. 7.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Lord, F.M., & Novick, M.R. (1968). *Statistical theories of mental test scores*. Reading, MA: Addison-Wesley Publishing.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Masters, G.N. (1982). A Rasch model for partial credit scoring. *Psychometrika*, 47, 149–174.

[Google Scholar](#) [WorldCat](#)

Mislevy, R.J., & Verhelst, N. (1990). Modeling item responses when different subjects employ different solution strategies. *Psychometrika*, 55, 195–216.

[Google Scholar](#) [WorldCat](#)

Muraki, E. (1990). Fitting a polytomous item response model to Likert-type data. *Applied Psychological Measurement*, 14, 59–71.

[Google Scholar](#) [WorldCat](#)

Muraki, E. (1992). A generalized partial credit model: Application of an EM algorithm. *Applied Psychological Measurement*, 16, 159–176.

[Google Scholar](#) [WorldCat](#)

Muraki, E., & Bock, R.D. (2003). *PARSCALE* (Version 4.1) [Computer Program]. Mooresville, IN: Scientific Software.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Muthén, L. K., & Muthén, B. O. (2007). *Mplus User's Guide*. Los Angeles, CA: Muthén & Muthén.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Parshall, C.G., Spray, J.A., Kalohn, J.C., & Davey, T. (2002). *Practical Considerations in Computerized-Based Testing*. New York: Springer.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Rasch, G. (1961). On general laws and the meaning of measurement in psychology. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability* (pp. 321–333). Berkeley, CA: University of California Press.

Rasch, G. (1980). *Probabilistic models for some intelligence and attainment tests*. Chicago, IL: University of Chicago Press. (Original work published 1960.)

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

R Development Core Team. (2007). *R: A language and environment for statistical computing* [Computer software]. Vienna, Austria:

p. 169 R Foundation for Statistical Computing. Program ↪ available at <http://www.r-project.org>. Last accessed July 16, 2012.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Reckase, M.D. (1989, Fall). Adaptive testing: The evolution of a good idea. *Educational Measurement: Issues and Practice*, 11–15.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Reckase, M.D. (2009). *Multidimensional Item Response Theory*. New York: Springer.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Rost, J. (1990). A logistic mixture distribution model for polychotomous item responses. *British Journal of Mathematical and Statistical Psychology*, 44, 75–92.

[Google Scholar](#) [WorldCat](#)

Samejima, F. (1969). Estimation of latent ability using a response pattern of graded scores. *Psychometrika Monograph Supplement*, No. 17.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Sands, W.A., Waters, B.K., & McBride, J.R. (Eds.) (1997). *Computerized Adaptive Testing: From Inquiry to Operation*. Washington, DC: American Psychological Association.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

SAS Institute (2002). *SAS for Windows: Version 9.1*. Carey, NC: Author.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Smith, E.V., & Kulikowich, J.M. (2004). An application of generalizability theory and many-facet Rasch measurement using a complex problem-solving skills assessment. *Educational and Psychological Measurement*, 64, 617–639.

[Google Scholar](#) [WorldCat](#)

Stocking, M. L., & Lord, F. M. (1983). Developing a common metric in Item Response Theory. *Applied Psychological Measurement*, 7, 201–210.

[Google Scholar](#) [WorldCat](#)

SYSTAT Software Incorporated (2007). *SYSTAT for Windows: Version 12*. San Jose, CA: Author.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Thissen, D.J., Chen, W.-H., & Bock, R.D. (2003). *MULTILOG* (Version 7.0) [Computer Program]. Mooresville, IN: Scientific Software.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Thurstone, L.L. (1925). A method of scaling psychological and educational tests. *The Journal of Educational Psychology*, 16, 433–451.

[Google Scholar](#) [WorldCat](#)

Tucker, L.R. (1946). Maximum validity of a test with equivalent items. *Psychometrika*, 11, 1–13.

[Google Scholar](#) [WorldCat](#)

Tuerlinckx, F., & De Boeck, P. (2001). The effect of ignoring item interactions on the estimated discrimination parameters in item response theory. *Psychological Methods*, 6, 181–195.

[Google Scholar](#) [WorldCat](#)

van der Linden, W.J., & Glass, C.A.W. (Eds.) (2010). *Elements of Adaptive Testing*. New York: Springer.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Von Davier, M. (2001). WINMIRA 2001 [Computer Program]. Kiel, Germany: *Institute für die Pädagogik der Naturwissenschaften an der Universität Kiel*. Available at www.winmira.von-davier.de/ Last accessed July 16, 2012.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Wainer, H., Bradlow, E.T., & Wang, X. (2007). *Testlet Response Theory*. New York: Cambridge University Press.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Wilson, M., & De Boeck, P. (2004). Descriptive and explanatory item response models. In De Boeck, P., & Wilson, M. (Eds.), *Explanatory Item Response Models: A generalized linear and nonlinear approach* (pp. 43–74). New York: Springer-Verlag.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Wright, B.D. (1977). Misunderstanding the Rasch Model. *Journal of Educational Measurement*, 14, 219–226.

[Google Scholar](#) [WorldCat](#)

Wu, M.L., Adams, R.J., & Wilson, M.R. (1997). *ConQuest: Multi-Aspect Test Software* [Computer Program]. Camberwell, Australia: Australian Council for Educational Research.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Yen, W. M. (1984). Effects of local item dependence on the fit and equating performance of the three-parameter logistic model. *Applied Psychological Measurement*, 8, 125–145.

[Google Scholar](#) [WorldCat](#)

Yen, W. M. (1993). Scaling Performance Assessments: Strategies for managing local item dependence. *Journal of Educational Measurement*, 30, 187–213.

[Google Scholar](#) [WorldCat](#)

Zimowski, M., Muraki, E., Mislevy, R.J., & Bock, R.D. (2003). *BILOG-MG* (Version 3.0) [Computer Program]. Mooresville, IN: Scientific

Software.

[Google Scholar](#)

[Google Preview](#)

[WorldCat](#)

[COPAC](#)

Zumbo, B. D. (1999). *A Handbook On The Theory And Methods Of Differential Item Functioning (DIF): Logistic Regression Modeling As A Unitary Framework For Binary And Likert-Type (Ordinal) Item Scores*. Ottawa, ON: Directorate of Human Resources Research and Evaluation, Department of National Defense.

[Google Scholar](#)

[Google Preview](#)

[WorldCat](#)

[COPAC](#)